

CHAPTER

11

CONSTRUCTION REQUIREMENTS FOR EXISTING BUILDINGS

General Comments

This chapter first appeared in the 2009 edition of the code, as Chapter 46. It brings many of the retroactive existing building fire protection and life safety requirements previously found in Chapters 7, 9 and 10 to one location for a more convenient and user-friendly enforcement tool. Requirements for premises identification of buildings remain in Chapter 5 and the existing building interior finish requirements remain in Chapter 8.

Excluding the general provisions and definitions, Chapter 11 is divided into five other sections. Section 1103 addresses fire safety requirements for existing buildings. This includes protection of vertical openings in shafts and openings for escalators. Section 1103 also contains provisions requiring automatic sprinkler, standpipe, and fire alarm and detection systems in existing buildings. Section 1104 specifies the minimum means of egress requirements for existing buildings and is based on materials once found in Chapter 10. Section 1105 establishes construction requirements for existing Group I-2 occupancies. Section 1106 sets forth the requirements for fire department access to outdoor storage of tires, which is retroactive to any existing storage conditions. Section 1107 requires a failure mode and effects analysis or another approved hazard mitigation analysis for existing energy storage systems that were installed prior to adoption the 2018 IFC or later editions.

Note that Section 1101.4 provides a level of reasonableness for enforcement of this chapter. It is not intended that existing conditions be immediately fixed, rather a process and timeline is determined between the building owner and fire code official.

Purpose

The purposes of this chapter are to provide minimum construction requirements for existing buildings and to provide them in one place. Many newer buildings may already meet these requirements and no further action will be necessary. Those that do not meet these minimal construction requirements will need to address those deficiencies. Interior finishes for existing buildings are dealt with separately in Chapter 8.

SECTION 1101—GENERAL

1101.1 Scope. The provisions of this chapter shall apply to existing buildings constructed prior to the adoption of this code.

C This section introduces Chapter 11 and indicates that it is applicable to all buildings constructed prior to the code's date of adoption.

1101.2 Intent. The intent of this chapter is to provide a minimum degree of fire and life safety to persons occupying existing buildings by providing minimum construction requirements where such existing buildings do not comply with the minimum requirements of the *International Building Code*.

C This chapter provides a single location for retroactive construction requirements for existing buildings. Included is a quick-reference table (see Table 1103.1) to determine whether there are any requirements that would be applicable as the inspector is conducting the inspection. Each provision in the code that refers to retroactive requirements now has a reference to this chapter.

Historically, there has been confusion as to when the construction-related requirements of this code are retroactive. Item 3 of Section 102.1 clarifies that the requirements of Chapter 11 apply to existing buildings.

Section 102.1, Item 3, also indicates that the retroactive construction provisions referred to are located in Chapter 11, thus clarifying which provisions in the code are actually construction provisions that should be applied to an existing building. Since only the provisions listed in this chapter would apply to an existing facility, all of the other construction provisions in the code are applicable to new construction. However, as has always been the case, the code official can still exercise judgment and declare a distinct hazard under Section 102.1, Item 4, for other items or operations not addressed in Chapter 11. See the commentary to Section 102.1 for further information.

Section 1104.1 makes it clear that existing buildings must still comply with the code under which the building was built or the minimum egress requirements in Section 1104, whichever is more restrictive. In this fashion, a building will not be allowed to reduce the egress system protection or design from the original approval. Section 1104 is not as restrictive as requirements for new construction and allows for the continued use of existing buildings when the egress is at an acceptable standard, but not in complete compliance with the *International Building Code*® (IBC®).

1101.3 Permits. Permits shall be required as set forth in Sections 105.5 and 105.6 and the *International Building Code*.

C See the commentary to Section 105.5 for operation permits and Section 105.6 for construction permits.

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1101.4 Owner notification. When a building is found to be in noncompliance with this chapter, the *fire code official* shall duly notify the *owner* of the building. Upon receipt of such notice, the *owner* shall, subject to the following time limits, take necessary actions to comply with the provisions of this chapter.

C See the commentary to Sections 113.3 and 113.3.2.

1101.4.1 Construction documents. *Construction documents* necessary to comply with this chapter shall be completed and submitted within a time schedule *approved* by the *fire code official*.

C See the commentary to Section 106.

1101.4.2 Completion of work. Work necessary to comply with this chapter shall be completed within a time schedule *approved* by the *fire code official*.

C See the commentary to Sections 105.3.1 and 105.2.3.

1101.4.3 Extension of time. The *fire code official* is authorized to grant necessary extensions of time where it can be shown that the specified time periods are not physically practical or pose an undue hardship. The granting of an extension of time for compliance shall be based on the showing of good cause and subject to the filing of an acceptable systematic plan of correction with the *fire code official*.

C See the commentary to Section 105.3.2.

SECTION 1102—DEFINITIONS

1102.1 Definitions. The following terms are defined in Chapter 2:

DUTCH DOOR.

EXISTING.

C Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2 or in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (see also commentary, Sections 201.1 through 201.4).

SECTION 1103—FIRE SAFETY REQUIREMENTS FOR EXISTING BUILDINGS

1103.1 Required construction. Existing buildings shall comply with not less than the minimum provisions specified in Table 1103.1 and as further enumerated in Sections 1103.2 through 1103.10.

The provisions of this chapter shall not be construed to allow the elimination of *fire protection systems* or a reduction in the level of fire safety provided in buildings constructed in accordance with previously adopted codes.

Exceptions:

1. Where a change in *fire-resistance rating* has been *approved* in accordance with Section 501.2 or 802.6 of the *International Existing Building Code*.
2. Group U occupancies.

C This section reinforces the provisions of Section 1101.1 and points to the table that provides a tabular summary of the requirements of this chapter and how they apply to the various occupancy groups. With respect to existing fire protection systems, this section reinforces the provisions of Section 901.4, which emphasizes the principle that systems installed and maintained in compliance with the codes and standards in effect at the time they were placed in service must remain operational at the same level at all times.

Exception 1 recognizes an allowance provided in the *International Existing Building Code*® (IEBC®). IEBC Sections 501.2 and 802.6 allow reductions in fire-resistance-rated construction if an automatic sprinkler system is added to a building. These reductions are required to be consistent with what is required in the IBC and also include extensive documentation and approval from the code official.

Exception 2 correlates with Table 1103.1. That table does not include provisions for Group U occupancies (as defined in Section 203 and IBC Section 312), which are generally miscellaneous structures that do not pose the same life safety hazard as other occupancy classifications.

TABLE 1103.1—OCCUPANCY AND USE REQUIREMENTS^a

SECTION	USE				OCCUPANCY CLASSIFICATION																			
	High-rise	Atrium or covered mall	Under-ground building	Tire storage	A	B	E	F	H-1	H-2	H-3	H-4	H-5	I-1	I-2	I-3	I-4	M	R-1	R-2	R-3	R-4	S	
1103.2	R	R	R	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	—	—	R	
1103.3	R	—	R	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	—	—	R	
1103.4.1	R	—	R	—	—	—	—	—	—	—	—	—	—	—	R	R	—	—	—	—	—	—	—	
1103.4.2	R	—	R	—	R	R	R	R	R	R	R	R	R	R	—	—	R	R	R	R	—	—	R	
1103.4.3	R	—	R	—	R	R	R	R	R	R	R	R	R	R	—	—	R	R	R	R	—	—	R	
1103.4.4	—	R	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
1103.4.5	—	—	—	—	—	R	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	
1103.4.6	—	—	—	—	R	—	R	R	R	R	R	R	R	R	R	R	R	—	R	R	R	R	R	
1103.4.7	—	—	—	—	R	—	R	R	R	R	R	R	R	R	R	R	R	—	R	R	R	R	R	
1103.4.8	R	—	R	—	R	R	R	R	R	R	R	R	R	R	—	—	R	R	R	R	R	R	R	
1103.4.9	R	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	—	—	—	
1103.4.10	—	—	—	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	
1103.5.1	—	—	—	—	R ^c	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
1103.5.2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	—	—	—	
1103.5.3	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R ^b	—	—	—	—	—	—	—	—	
1103.5.5	—	—	—	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	
1103.6.1	R	—	R	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	—	—	R	
1103.6.2	R	—	R	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	—	—	R	
1103.7.1	—	—	—	—	—	—	R	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
1103.7.2	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	—	—	—	—	
1103.7.3	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	—	—	—	
1103.7.4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	—	—	
1103.7.5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	
1103.7.6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	
1103.8	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	R	R	R	R	—	
1103.9	R	—	—	—	—	—	—	—	—	—	—	—	—	R	R	—	R	—	R	R	R	R	—	
1103.10	—	—	—	—	—	—	—	—	—	—	—	—	—	R	R	—	—	—	—	—	—	—	—	
1104	R	R	R	—	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	
1105	—	—	—	—	—	—	—	—	—	—	—	—	—	—	R	—	—	—	—	—	—	—	—	
1106	—	—	—	R	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	

R = The building is required to comply.

a. Existing buildings shall comply with the sections identified as “Required” (R) based on occupancy classification or use, or both, whichever is applicable.

b. Only applies to Group I-2, Condition 2 occupancies as established by the adopting ordinance or legislation of the jurisdiction.

c. Only applies to Group A-2 occupancies where alcoholic beverages are consumed.

1103.1.1 Historic buildings. Facilities designated as historic buildings shall develop a fire protection plan in accordance with NFPA 914. The fire protection plans shall comply with the maintenance and availability provisions in Sections 404.3 and 404.4.

C Section 102.6 generally does not require compliance with the code for historic buildings. However, that same section states that such buildings have a fire protection plan as required by this section. This plan will provide minimal ongoing safety for the occupants. This section also references back to the maintenance and availability requirements found in Chapter 4 to make sure the plans remain current and available to occupants and code officials.

1103.2 Emergency responder communications enhancement in existing buildings. Existing buildings other than Group R-3 that do not have *approved* in-building emergency response communications enhancement for emergency responders in the building based on existing coverage levels of the public safety communication systems, shall be equipped with such coverage according to one of the following:

1. Where an existing wired communication system cannot be repaired or is being replaced, or where not *approved* in accordance with Section 510.1, Exception 1.

2. Within a time frame established by the adopting authority.

Exception: Where it is determined by the *fire code official* that the in-building emergency responder communications enhancement system is not needed.

C This section specifies the requirements for in-building, emergency responder communication coverage in existing buildings other than Group R-3 occupancies. Group R-3 occupancies are essentially one- and two-family dwellings and such requirements would be excessive. Where emergency responder communication coverage is not adequate, Section 510.1, Exception 1 allows for the installation of a wired communication system. However, in existing buildings that have such a system or may have taken that exception, in the event the wired communication system becomes inoperative and cannot be repaired or is undergoing extended repair or replacement the exception provides the authority to the fire code official to determine if such a system is not necessary.

Where an existing building is determined to have inadequate coverage based on the requirements of Section 510.2, Item 2 of this section allows the fire code official to establish a time frame for compliance with Section 510.1. There are times when prior to construction it will be difficult to confidently assure that in-building, emergency response communication coverage will not be affected by the building itself. That is the reason these requirements are also applicable to existing facilities.

1103.3 Existing elevators. In other than Group R-3, existing elevators, escalators and moving walks shall comply with the requirements of Sections 1103.3.1 and 1103.3.2.

C The following two sections provide retroactive requirements that apply to existing elevators and, to a certain extent, escalators and moving walkways. Note that Section 604 also deals with the ongoing operation and maintenance of elevators in buildings. That section also has requirements related to fire service keys.

1103.3.1 Elevators, escalators and moving walks. Existing elevators, escalators and moving walks in Group I-2, Condition 2 occupancies and serving ambulatory care facilities shall comply with ASME A17.3.

C This section is specific to Group I-2, Condition 2 occupancies (hospitals) and ambulatory care facilities. Not only are existing elevators required to comply with ASME A17.3, but escalators and moving walks must also comply. Note that IBC Section 3001.3 would require ongoing maintenance of all elevators and conveying systems. This section specifies compliance with ASME A17.3, *Safety Code for Existing Elevators and Escalators*, for such maintenance.

The healthcare industry has historically been required to comply with regulations set forth by accreditation and certification agencies, such as The Joint Commission. ASME A17.3 has been referenced by guidelines adopted by The Joint Commission for over a decade and this requirement provides correlation of the code with the mandated healthcare industry standard.

The reference to ASME A17.3 requires that existing elevators, escalators, moving walks and their related operating equipment in Group I-2, Condition 2 occupancies and ambulatory care facilities to comply with a minimum level of safety. Because the occupants of these types of facilities are often incapable of self-preservation, it is essential to provide ASME-mandated features for occupant safety including escalator and moving walk emergency stop buttons and automatic skirt obstruction stop features.

1103.3.2 Elevator emergency operation. Existing elevators with a travel distance of 25 feet (7620 mm) or more above or below the main floor or other level of a building and intended to serve the needs of emergency personnel for firefighting or rescue purposes shall be provided with emergency operation in accordance with ASME A17.3.

Exceptions:

1. Buildings without occupied floors located more than 55 feet (16 764 mm) above or 25 feet (7620 mm) below the lowest level of fire department vehicle access where protected at the elevator shaft openings with additional fire doors in accordance with Section 716 of the *International Building Code* and where all of the following conditions are met:
 - 1.1. The doors shall be provided with vision panels of *approved* fire-protection-rated glazing so located as to furnish clear vision of the approach to the elevator. Such glazing shall not exceed 100 square inches (0.065 m²) in area.
 - 1.2. The doors shall be held open but be automatic-closing by activation of a fire alarm initiating device installed in accordance with the requirements of NFPA 72 as for Phase I Emergency Recall Operation, and shall be located at each floor served by the elevator; in the associated elevator machine room, control space, or control room; and in the elevator hoistway, where sprinklers are located in those hoistways.
 - 1.3. The doors, when closed, shall have signs visible from the approach area stating: "WHEN THESE DOORS ARE CLOSED OR IN CASE OF FIRE, ELEVATORS ARE OUT OF SERVICE. USE EXIT."
2. Buildings without occupied floors located more than 55 feet (16 764 mm) above or 25 feet (7620 mm) below the lowest level of fire department vehicle access where provided with *automatic sprinkler systems* installed in accordance with Section 903.3.1.1 or 903.3.1.2.

3. Freight elevators in buildings provided with both *automatic sprinkler systems* installed in accordance with Section 903.3.1.1 or 903.3.1.2 and not less than one ASME 17.3-compliant elevator serving the same floors.

Elimination of previously installed Phase I emergency recall or Phase II emergency in-car systems shall not be permitted.

C This section establishes requirements for existing elevators. Existing elevators that travel 25 feet (7620 mm) or more above or below the main level must, as a minimum, be equipped with emergency operation capabilities that comply with ASME A17.3. New elevator installations are held to more restrictive guidelines for increased cost effectiveness and must have both emergency recall (Phase I) and emergency in-car operation (Phase II) systems to comply with ASME A17.1 for any amount of travel distance, as required by Section 604.2. The ASME standards are safety codes for elevators and escalators: ASME A17.3 for existing elevators and ASME A17.1 for new elevator installations. Automatic fire detection devices provided for Phase I emergency recall operation must be installed in accordance with Section 907.3.3.

There are several exceptions based on the expense of providing Phase I recall and Phase II emergency operation for existing elevators. ASME A17.3, Rule 211.3 (Firefighters' Service—Automatic Elevators) states, "All automatic (nondesignated attendant) operation elevators having a travel of 25 ft (7.62 m) or more above or below the designated level shall conform to the requirements of this Rule."

These changes are needed because of the excessive expense of compliance for many of these retrofits. Application of the mandatory Phase I and Phase II retrofits required by A17.3 triggers complete replacement of the elevator machinery at costs running into the hundreds of thousands of dollars. What is a relatively inexpensive fire safety feature in a new elevator installation can be unreasonably onerous when applied to existing elevators. It is therefore reasonable to codify alternate methods for building owners to meet the intent of the codes. It is important to note that the 25-foot travel threshold for an ASME A17.3 emergency operation retrofit is not mitigated by occupant load; number of stories; elevator use; building fire or smoke compartment conditions; the presence of sprinklers; or any building-specific operational firefighting considerations. For elevators installed prior to the adoption of newer elevator emergency operations features, many existing three-story buildings require retrofits: elevators in nonatmospherically segregated spaces such as low-rise atriums require retrofits; vintage freight elevators, regardless of maintenance condition or the ability of building occupants to access them, require retrofits. The problem of inflexible compliance options is magnified by the division of professional jurisdictions because elevator authorities typically do not have the expertise to assess fire risk on a total building basis; hence there is a need for the code to scope retroactive fire safety provisions.

By allowing for more affordable or effective alternatives to the Phase I and Phase II retrofits, less opposition should exist to adoption of ASME A17.3 and its other retroactive safety requirements such as safety bulkheads for hydraulic elevators and door restrictors, thereby enhancing overall elevator safety.

Phase I emergency recall is intended to prevent elevator users from being discharged at a floor that is engaged in a fire; to prevent occupants from accessing an elevator during a fire; and, in a detected fire condition, to return the elevator car to a designated floor for firefighter access for operations and rescue.

Phase II emergency in-car operation is intended to provide firefighters the ability to operate the elevator for firefighting operations.

Exception 1 meets the intent of the code by:

- Providing an additional fire door, with a vision panel, between elevator occupants and a fire engaged floor. This protects occupants from car and hoistway doors automatically opening directly to a fire event. The vision panel lets occupants view fire risk and select another floor to travel to for egress. This door provides an additional barrier to smoke and water contamination of the hoistway and improves building compartmentation.
- Providing additional signs on the added door immediately at the elevator opening telling building occupants to not use the elevator when the door is closed. In other words, if you can read the sign you shouldn't use the elevator.
- Recognizing that common firefighting operations policy requires firefighters to use the stairs to address any fire on the sixth floor or lower in a building; meaning that in low-rise buildings immediate firefighter access to the elevators is not as critical. The sixth floor of a building typically corresponds to the code's definition of a high-rise building as one with "an occupied floor located more than 75 feet (22 860 mm) above the lowest level of fire department vehicle access." This is broadly recognized as the elevation at which a compromise is achieved between the inherent hazard of using an elevator in a structural fire response and the need to respond in a timely way to the fire.

Exception 1 also tacitly acknowledges that these elevators were never "intended to serve the needs of emergency personnel for firefighting or rescue purposes," as the charging section states.

Locations of fire alarm-initiating devices used to initiate door closing are identical to those used to initiate Phase I operations via reference to NFPA 72 by ASME A17.3.

Exception 2 recognizes that there is no documented loss of life while in an elevator during a structural fire in a building protected with automatic fire sprinklers. It is reasonable that if an owner has already provided superior fire safety features in an existing building he or she will not be required to retrofit expensive fire safety features of limited value.

For buildings otherwise within the scope of the exception, Exception 2 provides an incentive to provide fire sprinklers.

The references to Sections 903.3.1.1 and 903.3.1.2 mean that only a fire sprinkler system installed in accordance with the appropriate technical standard (NFPA 13 or NFPA 13R) will permit relaxation of the Phase I and Phase II emergency operations requirements.

Fifty-five feet was selected as the upper limit for the floor height of buildings for a couple of reasons. It was suggested that a height limit of roughly four stories would be preferable when considering response factors associated with ascertaining elevator occupant locations and conditions. Since 55 feet is the elevation at which automatic sprinkler protection is triggered, and since it roughly corresponds to four stories, it is offered as a suitable threshold.

Exception 3 recognizes that buildings having another elevator equipped with Phase I and Phase II emergency operations already meet the intent of Phase II firefighter operations provisions. The fire sprinkler requirement and the limited passenger use of freight elevators means that the likelihood of life loss in the freight elevator is very low. For buildings without fire sprinklers, this provision provides an incentive to install fire sprinkler systems.

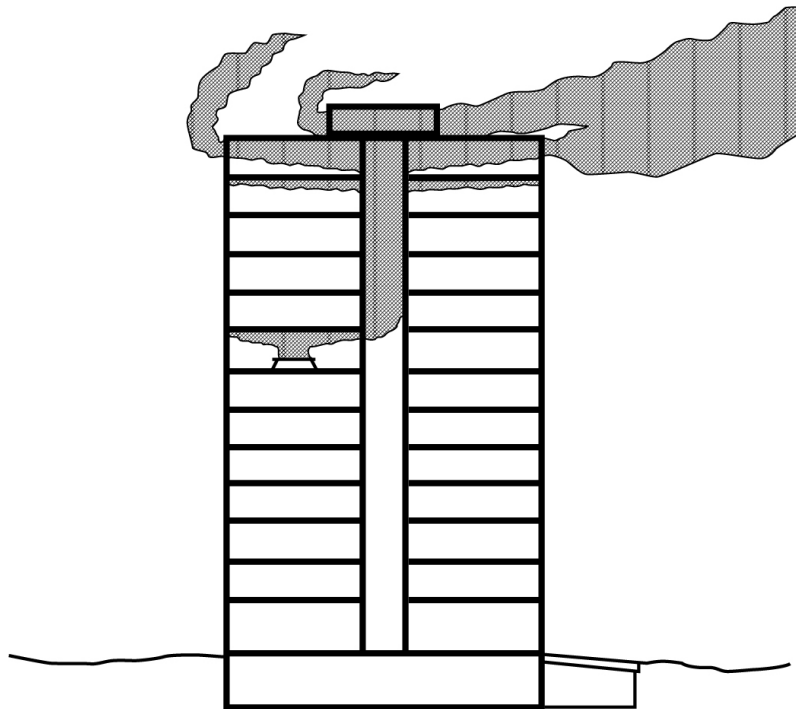
The final paragraph of this section clarifies that previously installed Phase I emergency recall and Phase II emergency in-car operations are not to be removed.

1103.4 Vertical openings. Interior vertical openings, including but not limited to *stairways*, elevator hoistways, service and utility shafts, that connect two or more stories of a building shall be enclosed or protected as specified in Sections 1103.4.1 through 1103.4.10.

C Vertical openings that are not properly protected can act as a chimney for smoke, hot gases and products of combustion. Unprotected floor openings have been a major contributing factor in many large loss-of-life fires. This section retroactively requires the enclosure of vertical openings between floors with approved fire barriers. The intent is to increase the level of safety in all buildings by enclosing unprotected floor openings and, thus, removing the avenue for unimpeded fire and smoke spread. In particular, the code is intended to ensure that means of egress in existing multiple-story buildings are given a reasonable level of protection.

Vertical communication between floors (including stairways, escalator openings, elevator hoistways, trash and laundry chutes, and other types of building service and mechanical shafts) can contribute significantly to fire and smoke spread because of “stack effect” (see Commentary Figure 1103.4). This section and Sections 1103.4.1 through 1103.4.10 require that interior shafts be enclosed with fire barriers of the fire-resistance rating indicated. These assemblies must be constructed and protected as required by the IBC. The allowance for sprinklers in the various sections recognizes the value of automatic sprinkler systems conforming to Section 903.3.1.1 (NFPA 13 system) or 903.3.1.2 (NFPA 13R system), as applicable.

Commentary Figure 1103.4—Stack Effect and Smoke Migration via Vertical Shafts



1103.4.1 Group I-2 and I-3 occupancies. In Group I-2 and I-3 occupancies, interior vertical openings connecting two or more stories shall be protected with 1-hour *fire-resistance-rated* construction.

Exceptions:

1. In Group I-2, unenclosed vertical openings not exceeding two connected stories and not concealed within the building construction shall be permitted as follows:
 - 1.1. The unenclosed vertical openings shall be separated from other unenclosed vertical openings serving other floors by a *smoke barrier*.
 - 1.2. The unenclosed vertical openings shall be separated from *corridors* by smoke partitions.
 - 1.3. The unenclosed vertical openings shall be separated from other fire or *smoke compartments* on the same floors by a *smoke barrier*.
 - 1.4. On other than the lowest level, the unenclosed vertical openings shall not serve as a required *means of egress*.
2. In Group I-2, atriums connecting three or more stories shall not require 1-hour *fire-resistance-rated* construction where the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3, and all of the following conditions are met:
 - 2.1. For other than existing *approved* atriums with a smoke control system, where the atrium was constructed and is maintained in accordance with the code in effect at the time the atrium was created, the atrium shall have a smoke control system that is in compliance with Section 909.
 - 2.2. Glass walls forming a smoke partition or a glass-block wall assembly shall be permitted where in compliance with Condition 2.2.1 or 2.2.2.
 - 2.2.1. Glass walls forming a smoke partition shall be permitted where all of the following conditions are met:
 - 2.2.1.1. Automatic sprinklers are provided along both sides of the separation wall and doors, or on the room side only if there is not a walkway or occupied space on the atrium side.
 - 2.2.1.2. The sprinklers shall be not more than 12 inches (305 mm) away from the face of the glass and at intervals along the glass of not greater than 72 inches (1829 mm).
 - 2.2.1.3. Windows in the glass wall shall be nonoperating type.
 - 2.2.1.4. The glass wall and windows shall be installed in a gasketed frame in a manner that the framing system deflects without breaking (loading) the glass before the *automatic sprinkler system* operates.
 - 2.2.1.5. The sprinkler system shall be designed so that the entire surface of the glass is wet upon activation of the *automatic sprinkler system* without obstruction.
 - 2.2.2. A *fire barrier* is not required where a glass-block wall assembly complying with Section 2110 of the *International Building Code* and having a $\frac{3}{4}$ -hour *fire protection rating* is provided.
 - 2.3. Where doors are provided in the glass wall, they shall be either self-closing or automatic-closing and shall be constructed to resist the passage of smoke.
3. In Group I-3 occupancies, exit *stairways* or *ramps* and *exit access stairways* or *ramps* constructed in accordance with Section 408 of the *International Building Code*.

C This section specifically addresses vertical openings only for Group I-2 and I-3 occupancies. In nursing homes, hospitals and jails, the first response to a fire is to defend in place or move residents, patients or detainees to an adjacent smoke compartment. To maintain the separation needed to limit the spread of smoke, an increased level of protection is required. There are three exceptions to the requirement for 1-hour enclosure of vertical openings (for vertical openings in other occupancies, see Section 1103.4.8).

Exceptions 1 and 2 are applicable for Group I-2 occupancies (nursing homes and hospitals). Exceptions 1 and 2 are intended to more appropriately address floor openings in existing construction. Without these exceptions, the IBC would allow construction of a floor opening without a 1-hour fire barrier in certain specific cases. This also impacts all historical nonrated floor openings that have been reviewed, approved and maintained. Unrated vertical openings had been allowed in hospitals and nursing homes previously. Atriums have been installed with various types of smoke venting and removal systems over the past few decades. These exceptions attempt, while maintaining an appropriate level of protection, to address general requirements that have been broadly used throughout older editions of the building codes. These exceptions are consistent with federal regulations governing hospitals and nursing homes. It should be remembered that these provisions are retroactive. They apply regardless of whether an alteration is to be undertaken.

Exception 3 provides consistency with Exception 1 to Section 1023.2, which allows interior exit stairways in Group I-3 occupancies to be in accordance with IBC Section 408.3.8.

Being able to see into and out of a stairway enclosure is an important security feature.

1103.4.2 Three to five stories. In other than Group I-2 and I-3 occupancies, interior vertical openings connecting three to five stories shall be protected by either 1-hour *fire-resistance-rated* construction or an *automatic sprinkler system* shall be installed throughout the building in accordance with Section 903.3.1.1 or 903.3.1.2.

Exceptions:

1. Vertical opening protection is not required for Group R-3 occupancies.
2. Vertical opening protection is not required for open parking garages.
3. Vertical opening protection for escalators shall be in accordance with Section 1103.4.5, 1103.4.6 or 1103.4.7.
4. *Exit access stairways* and *ramps* shall be in accordance with Section 1103.4.8.

C See the commentary to Section 1103.4. The exceptions are consistent with the exceptions to the shaft enclosure requirements of the IBC. Exception 1 exempts single-family dwellings and small congregate residences from vertical opening enclosures because of occupant familiarity with the premises and the relatively small occupant load. Exception 2 recognizes that it would be impractical and unnecessary to enclose shafts in parking occupancies where open ramps are a fundamental design element. Exception 3 addresses escalator openings and refers to the applicable code sections that apply. Exception 4 recognizes that exit access stairways and ramps have various allowances that would not require enclosure. These allowances are consistent with Chapters 7 of the IBC and 10 of this code (see Section 1103.4.8).

1103.4.3 More than five stories. In other than Group I-2 and I-3 occupancies, interior vertical openings connecting more than five stories shall be protected by 1-hour *fire-resistance-rated* construction.

Exceptions:

1. Vertical opening protection is not required for Group R-3 occupancies.
2. Vertical opening protection is not required for open parking garages.
3. Vertical opening protection for escalators shall be in accordance with Section 1103.4.5, 1103.4.6 or 1103.4.7.
4. *Exit access stairways* and *ramps* shall be in accordance with Section 1103.4.8.

C See the commentary to Section 1103.4. The exceptions are consistent with the exceptions to the shaft enclosure requirements of the IBC. Exception 1 exempts single-family dwellings and small congregate residences from vertical opening enclosures because of occupant familiarity with the premises and the relatively small occupant load. Exception 2 recognizes that it would be impractical and unnecessary to enclose shafts in parking occupancies where open ramps are a fundamental design element. Exception 3 addresses escalator openings and refers to the applicable code sections that apply. Exception 4 recognizes that exit access stairways and ramps have various allowances that would not require enclosure. These allowances are consistent with Chapters 7 of the IBC and 10 of this code (see Section 1103.4.8).

1103.4.4 Atriums and covered malls. In other than Group I-2 and I-3 occupancies, interior vertical openings in a covered mall building or a building with an atrium shall be protected by either 1-hour *fire-resistance-rated* construction or an *automatic sprinkler system* shall be installed throughout the building in accordance with Section 903.3.1.1 or 903.3.1.2.

Exceptions:

1. Vertical opening protection is not required for Group R-3 occupancies.
2. Vertical opening protection is not required for open parking garages.
3. *Exit access stairways* and *ramps* shall be in accordance with Section 1103.4.8.

C See the commentary to Section 1103.4. The exceptions are consistent with the allowances for atrium and mall enclosure requirements of the IBC. Exception 1 exempts single-family dwellings and small congregate residences from vertical opening enclosures because of occupant familiarity with the premises and the relatively small occupant load. Exception 2 recognizes that it would be impractical and unnecessary to enclose shafts in parking occupancies where open ramps are a fundamental design element. Exception 3 recognizes that exit access stairways and ramps have various allowances that would not require enclosure. These allowances are consistent with Chapter 7 of the IBC, Chapter 10 of this code, and Sections 402 and 404 of the IBC. Escalators within a mall are addressed in Sections 1103.4.5, 1103.4.6 and 1103.4.7 (see Section 1103.4.8).

1103.4.5 Escalators in Group B and M occupancies. In Group B and M occupancies, escalators creating vertical openings connecting any number of stories shall be protected by either 1-hour *fire-resistance-rated* construction or an *automatic sprinkler system* in accordance with Section 903.3.1.1 installed throughout the building, with a draft curtain and closely spaced sprinklers around the escalator opening.

C This section addresses escalator openings in Group B and M occupancies and provides the option of either enclosing the vertical opening with 1-hour construction or equipping the building throughout with an NFPA 13 sprinkler system with the vertical opening protected by draft curtains and closely spaced sprinklers. Section 9.3.5.2 of NFPA 13 provides specifications on the construction and installation of the draft curtains ("draftstops" in NFPA 13) and Section 9.3.5.3 provides installation criteria for the automatic sprinklers for protecting the opening. There is no limit to the number of stories connected by vertical openings protected by either option in this section. IBC Section 712.1.3 contains similar, but more stringent, requirements for escalator openings in new buildings.

1103.4.6 Escalators connecting four or fewer stories. In other than Group B and M occupancies, escalators creating vertical openings connecting four or fewer stories shall be protected by either 1-hour *fire-resistance-rated* construction or an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 shall be installed throughout the building, and a draft curtain with closely spaced sprinklers shall be installed around the escalator opening.

C In other than Group B and M occupancies, escalators creating vertical openings connecting four or fewer stories shall either be protected by 1-hour fire-resistance-rated construction or an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2 shall be installed throughout the building and a draft curtain with closely spaced sprinklers shall be installed around the escalator opening.

This section addresses escalator openings connecting a maximum of four stories in other than Group B and M occupancies. Similar to Section 1103.4.5, it provides the option of either enclosing the vertical opening with 1-hour construction or equipping the building throughout with an NFPA 13 or 13R sprinkler system with the vertical opening protected by draft curtains (draftstops) and closely spaced sprinklers. Section 9.3.5.2 of NFPA 13 provides specifications on the construction and installation of the draft curtains (“draftstops” in NFPA 13) and Section 9.3.5.3. provides installation criteria for the automatic sprinklers for protecting the opening (also see commentary, Section 1103.4). IBC Section 712.1.3.1 contains similar, but more stringent, requirements for escalator openings in new buildings.

1103.4.7 Escalators connecting more than four stories. In other than Group B and M occupancies, escalators creating vertical openings connecting five or more stories shall be protected by 1-hour *fire-resistance-rated* construction.

C This section addresses escalator openings connecting five stories or more in other than Group B and M occupancies and provides no option to enclosing the vertical opening with 1-hour construction (see commentary, Section 1103.4). IBC Section 712.1.3 contains similar, but more stringent, requirements for escalator openings in new buildings.

1103.4.8 Occupancies other than Groups I-2 and I-3. In other than Group I-2 and I-3 occupancies, floor openings containing *exit access stairways* or *ramps* that do not comply with one of the conditions listed in this section shall be protected by 1-hour *fire-resistance-rated* construction.

1. *Exit access stairways* and *ramps* that serve, or atmospherically communicate between, only two stories. Such interconnected stories shall not be open to other stories.
2. In Group R-1, R-2 or R-3 occupancies, *exit access stairways* and *ramps* connecting four stories or less serving and contained within an individual *dwelling unit* or *sleeping unit* or live/work unit.
3. *Exit access stairways* and *ramps* in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, where the area of the vertical opening between stories does not exceed twice the horizontal projected area of the *stairway* or *ramp*, and the opening is protected by a draft curtain and closely spaced sprinklers in accordance with NFPA 13. In other than Group B and M occupancies, this provision is limited to openings that do not connect more than four stories.
4. *Exit access stairways* and *ramps* within an atrium complying with the provisions of Section 404 of the *International Building Code*.
5. *Exit access stairways* and *ramps* in open parking garages that serve only the parking garage.
6. *Exit access stairways* and *ramps* serving open-air seating complying with the *exit access* travel distance requirements of Section 1030.7 of the *International Building Code*.
7. *Exit access stairways* and *ramps* serving the balcony, gallery or press box and the main assembly floor in occupancies such as theaters, *places of religious worship*, auditoriums and sports facilities.

C Vertical openings in Group I-2 and I-3 occupancies are specifically addressed in Section 1103.4.1. This section addresses vertical openings in other occupancies. There are many provisions for new construction in Chapter 10 that permit exit access stair floor openings without a fire-rated enclosure. Without this section, many of these permitted floor openings would be required to be fire rated by this chapter. In addition, a building constructed under this code in full compliance with Chapter 10 would be in violation of Sections 1103.4 through 1103.4.8 upon issuance of the certificate of occupancy if these allowances were not provided. Clearly it was not the intent for Chapter 11 to contradict Chapter 10. This section notes all of the current conditions that allow an unenclosed exit access stair or ramp. All of these conditions originate from Section 1019. It is important to note that this section and these conditions only apply to exit access stairways and ramps. This section does not apply to exit stairways. Exit stairways are not exempted from enclosure.

1103.4.9 Waste and linen chutes. In Group I-2 occupancies, existing waste and linen chutes shall comply with Sections 1103.4.9.1 through 1103.4.9.5.

C This section is intended to clarify the allowable use and construction of chutes and incinerators in Group I-2 occupancies (hospitals and nursing homes). These items are used as integral parts of the operation of a healthcare facility, especially the waste or linen chutes. Some incinerators are still in use, but the requirements found in Sections 1103.4.9.1 through 1103.4.9.5 seek to separate them from other vertical openings, especially a trash chute, by requiring a separate discharge room from the incinerator. Waste and linen chutes in other occupancies are addressed in Section 1103.4.8.

1103.4.9.1 Enclosure. Chutes shall be enclosed with 1-hour *fire-resistance-rated* construction. Opening protectives shall be in accordance with Section 716 of the *International Building Code* and have a *fire protection rating* of not less than 1 hour.

C At minimum, chutes must be enclosed with 1-hour fire-resistance-rated construction. This is consistent with IBC Section 713.4 for shaft enclosures for chutes connecting three stories or less. Openings other than chute access doors addressed in Section 1103.4.9.2 must meet IBC Section 716 requirements.

1103.4.9.2 Chute intakes. Chute intakes shall comply with Section 1103.4.9.2.1 or 1103.4.9.2.2.

C This section focuses on where chutes can be accessed throughout a building. There are two configurations addressed. The first, in Section 1103.4.9.2.1, addresses access from the corridor. Section 1103.4.9.2.2 then addresses chute intakes accessed from a chute intake room.

1103.4.9.2.1 Chute intake direct from corridor. Where intake to chutes is direct from a *corridor*, the intake opening shall be equipped with a chute-intake door in accordance with Section 716 of the *International Building Code* and having a *fire protection rating* of not less than 1 hour.

C In facilities that predate current requirements, some chute doors open into corridors. As it is not practical to reconstruct such chutes to meet modern standards, this section aims to directly address that situation by defining requirements for the chutes' safe maintenance. This configuration has a more direct effect on the safety of the occupants than those chute intakes addressed in Section 1103.4.9.2.2. Therefore, the opening must be in accordance with IBC Section 716 and be rated for at least 1 hour.

1103.4.9.2.2 Chute intake via a chute-intake room. Where the intake to chutes is accessed through a chute-intake room, the room shall be enclosed with 1-hour *fire-resistance-rated* construction. Opening protectives for the intake room shall be in accordance with Section 716 of the *International Building Code* and have a *fire protection rating* of not less than $\frac{3}{4}$ hour. Opening protectives for the chute enclosure shall be in accordance with Section 1103.4.9.1.

C This configuration provides a layer of protection to the building occupants by first entering a room that is protected with 1-hour fire-resistance-rated construction. The door rating is only required to be $\frac{3}{4}$ hour versus 1 hour as required in Section 1103.4.9.2.1. The door for the chute intake itself would still be required to be 1 hour.

1103.4.9.3 Automatic sprinkler system. Chutes shall be equipped with an *approved automatic sprinkler system* in accordance with Section 903.2.11.2.

C This section would retroactively require compliance with Section 903.2.11.2 for sprinkler placement within the chute.

1103.4.9.4 Chute discharge rooms. Chutes shall terminate in a dedicated chute discharge room. Such rooms shall be separated from the remainder of the building by not less than 1-hour *fire-resistance-rated* construction. Opening protectives shall be in accordance with Section 716 of the *International Building Code* and have a *fire protection rating* of not less than 1 hour.

C This section provides minimal requirements for protection of the chute discharge room. This provides a minimal separation requirement for existing chute discharge rooms. This is consistent with the requirements in IBC Section 713.13.4 for shafts connecting three stories or less in new construction. The rating of the door must also be 1 hour.

1103.4.9.5 Chute discharge protection. Chute discharges shall be equipped with a self-closing or automatic-closing opening protective in accordance with Section 716 of the *International Building Code* and having a *fire protection rating* of not less than 1 hour.

C This section requires that the chute discharge room continue the protection of the enclosure through opening protection. The discharge must also have a 1-hour fire protection rating and be self-closing or automatic-closing. This is consistent with the requirement of IBC Section 713.13.4 that the opening protection have the same rating as the enclosure. The opening must also be self-closing or automatic-closing.

1103.4.10 Flue-fed incinerators. Existing flue-fed incinerator rooms and associated flue shafts shall be protected with 1-hour *fire-resistance-rated* construction and shall not have other vertical openings connected with the space other than the associated flue. Opening protectives shall be in accordance with Section 716 of the *International Building Code* and have a *fire protection rating* of not less than 1 hour.

C Because of regulation from the EPA and other entities, flue-fed incinerators are frequently found abandoned or otherwise unused in existing buildings. This requirement is intended to separate and protect any potential hazard of inactive incinerator systems from the rest of the building.

1103.5 Sprinkler systems. An *automatic sprinkler system* shall be provided in existing buildings in accordance with Sections 1103.5.1 through 1103.5.5.

C This section introduces the retroactive sprinkler requirements of Sections 1103.5.1 through 1103.5.5.

1103.5.1 Group A-2. Where alcoholic beverages are consumed in a Group A-2 occupancy having an *occupant load* of 300 or more, the *fire area* containing the Group A-2 occupancy shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1.

C This requirement implements Recommendation 1 of the NIST *Report of the Technical Investigation of The Station Nightclub Fire* (NIST NCSTAR 2), Volume I. Recommendation 1 states: “Model codes should require sprinkler systems for all new and existing nightclubs regardless of size.” Group A-2 occupancies involve conditions such as large occupant loads, high occupant density, significant fuel loading and moveable furnishings and decorations. Group A-2 occupancies also include the potential for reduced lighting levels, high noise levels, combustible decorations, strobe and flashing lights, alcohol consumption, and confusing egress paths. Each of these alone can be a significant issue, but when combined they lead to the inability of the occupants to promptly and safely exit the building under fire conditions.

This proposal does not reach as far as the recommendation from NIST. While the NIST proposal recommends fire sprinklers in all facilities, this section requires the existing Group A-2 occupancy fire area with an occupant load in excess of 300 where alcoholic drinks are consumed to be equipped with an automatic sprinkler system. The section does not require fire areas containing A-2 occupancies less than 300 to be protected, nor does it require the entire floor to be protected. Setting the threshold at 300 occupants will place the requirement where the higher potential for loss of life exists.

Additionally, this section focuses on Group A-2 occupancies that serve alcoholic beverages. This is felt to be more in line with the NIST recommendations that were made following The Station nightclub fire, recognizing that the intoxication of patrons plays a significant role in the potential risk of injury or loss of life in the event of a fire. In addition, limiting the scope of the change to only those occupancies where alcoholic beverages are consumed allows a connection to licensing laws that jurisdictions typically have in place for the sale of such beverages. Such licensing laws, where they apply, will provide significant leverage for jurisdictions to be able to effectively enforce the requirement for an automatic sprinkler system as a condition of being code compliant and issuance of a license.

Note that Section 903.2.1, which addresses new Group A-2 occupancies, requires an automatic sprinkler system throughout the story containing the group A-2 occupancy and all stories below where the fire area exceeds 5,000 square feet, has an occupant load of 100 or more, or is located on a story other than the level of exit discharge. Installing automatic sprinklers in a new building is much more efficient and cost-effective. The automatic sprinkler requirements for new Group A-2 occupancies include anything that qualifies as Group A-2, whether or not alcohol is served. This is more reasonable to ask of new construction. In existing construction, this section is aimed at the higher-risk Group A-2 occupancies in recognition of the difficulty in requiring sprinklers in existing buildings.

1103.5.2 Group I-2. In Group I-2, an *automatic sprinkler system* shall be provided in accordance with Section 1105.9.

C This section is merely a pointer to the sprinkler requirements for Group I-2 occupancies found in Section 1105.9. Section 1105.9 requires that an automatic sprinkler system be provided on the floor containing the Group I-2 occupancy, all floors between that level and the level of exit discharge, and all floors below the level of exit discharge. This requirement is less restrictive than what is found in Section 1103.5.3, since Section 1103.5.3 requires the entire building to be equipped throughout with an automatic sprinkler system. Note that I-2 occupancies include nursing homes and hospitals. Section 1103.5.3 only addresses hospitals.

1103.5.3 Group I-2, Condition 2. In addition to the requirements of Section 1103.5.2, existing buildings of Group I-2, Condition 2 occupancy shall be equipped throughout with an *approved automatic sprinkler system* in accordance with Section 903.3.1.1. The *automatic sprinkler system* shall be installed as established by the adopting ordinance. [DATE BY WHICH SPRINKLER SYSTEM MUST BE INSTALLED].

C This section is specific to hospitals and is more restrictive than the requirements of Sections 1103.5.2 and 1105.9. Instead of requiring the installation of an automatic sprinkler system from the story containing the Group I-2, Condition 2 occupancy to the level of exit discharge and below the level of exit discharge, this section requires the building to be equipped throughout with an automatic sprinkler system. This requirement, however, is based on the adopting ordinance establishing a date by which the automatic sprinkler system will be installed. Sprinkler systems are a vital safety system that protects building systems and components.

To ensure continuous operation in healthcare facilities, the installation of automatic sprinkler systems needs to be carefully planned so as to not adversely affect care recipient health. Accessing and exposing ceiling spaces can create conditions leading to infection and possibly death to care recipients with compromised immune systems. In many situations, hospitals may not be able to appropriately retrofit the installation of a fire suppression system; in those situations, a time frame is needed to replace facilities. The period for adoption of this proposed requirement has been left to the local authority having jurisdiction. Coordinating the time frame for adoption with federal requirements is recommended. It is currently anticipated that the Centers for Medicaid and Medicare (the federal authority having jurisdiction) will require retroactive installation of automatic sprinkler systems in hospitals by the year 2021. However, the exact time frame was uncertain at the time of development of this requirement. Facilities will require lead time to meet this

standard without adversely affecting the health of patients and disrupting patient care. These are the same factors that a jurisdiction should consider when choosing a date for adoption.

It should be also clear that this change is a separate measure that must be taken in addition to the current requirement. It is not intended to allow a facility to have a time frame for installing the current requirement (although jurisdictions may choose to do this). Nor is it intended to imply that the entire building containing a hospital should install an automatic sprinkler system immediately. At a minimum, a 3-year time frame is recommended for implementation of this requirement. This considers the process planning, capital approval, regulatory approval, design and installation of the sprinkler system. The capital-planning piece of a large-scale initiative such as a building-wide sprinkler system, normally spans multiple fiscal years.

1103.5.4 High-rise buildings. Where Appendix M has not been adopted, existing high-rise buildings that do not have a previously approved automatic sprinkler system shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1 where any of the following conditions apply:

1. The high-rise building has an occupied floor located more than 120 feet (36 576 mm) above the lowest level of fire department vehicle access.
2. The high-rise building has occupied floors located more than 75 feet (22 860 mm) and not more than 120 feet (36 576 mm) above the lowest level of fire department vehicle access, and the building does not have at least two *interior exit stairways* complying with Section 1104.10 that are separated from the building interior by fire assemblies having a *fire-resistance rating* of not less than 2 hours with opening protection in accordance with Table 716.1(2) of the *International Building Code*.
3. The high-rise building has occupied floors located more than 75 feet (22 860 mm) and not more than 120 feet (36 576 mm) above the lowest level of fire department vehicle access, and the building does not have a *fire alarm system* that includes smoke detection in mechanical equipment, electrical, transformer, telephone equipment and similar rooms; *corridors*; elevator lobbies; and at doors penetrating *interior exit stairway* enclosures.

Building owners shall file a compliance schedule with the *fire code official* not later than 365 days after receipt of a written notice. The compliance schedule shall not exceed 12 years for completion of the *automatic sprinkler system* retrofit.

C Nonsprinklered high-rise buildings represent a significant risk to occupants and emergency responders, and the risk increases as a function of building height and where a building has insufficient means of egress and inadequate fire detection systems. These risks have been well documented over time in the news media and NFPA reports (<https://www.nfpa.org/-/media/Files/News-and-Research/Fire-statistics-and-reports/Building-and-life-safety/oshighrise.pdf>), and while catastrophic fires may be infrequent, Grenfell Tower provided another reminder of the potential consequences. This section is intended to compliment Appendix M, which is an optional appendix that, if adopted by a jurisdiction, requires all high-rise buildings to be retrofitted with automatic sprinklers. For jurisdictions not adopting Appendix M, this section provides a less stringent retrofit requirement that offers an incremental step between retrofitting all high-rise buildings with sprinklers versus none at all. High-rise buildings meeting one of three specified conditions would be impacted:

1. Buildings having occupied floors located more than 120 feet above the lowest level of fire department access were selected because these buildings are well beyond the capability of most fire departments to handle with an exterior attack; represent an extreme difficulty with respect to mustering resources on upper floors for interior fire attack; and are difficult to evacuate, particularly when mobility challenged occupants are involved. The 120-foot threshold correlates with IFC Section 907.2.13.3, which requires multiple-channel voice evacuation systems for buildings, and IBC Section 403.6.1, which requires fire service access elevators. Both sections address buildings with an occupied floor more than 120 feet above the lowest level of fire department vehicle access.
2. Any nonsprinklered high-rise building that does not have at least two exit stairs enclosed by a 2-hour fire-rated assembly should be retrofitted with sprinklers. The challenge of occupant evacuation with the counter-flow of emergency responders taking equipment to the fire floor, particularly when mobility-challenged occupants must be brought down stairs, is sufficient to justify the need for sprinklers. And if the stairs are compromised and charged with smoke, occupants cannot be evacuated and firefighter access is further hampered by low visibility and the need to use SCBA while climbing to a fire floor. The 2-hour rating is justified as a minimum requirement because firefighters may need to perform an extended attack in a nonsprinklered building, as has been demonstrated in fires such as the First Interstate Bank Fire in Los Angeles.
3. Any nonsprinklered high-rise building that does not have a reasonable level of automatic fire detection in common areas to detect a fire, warn occupants and contact emergency responders should be retrofitted with sprinklers. Fires grow at an exponential rate, and delayed detection, alarm and emergency response significantly increases the risk to occupants and emergency responders. It is simply common sense that nonsprinklered high-rise buildings should have a reasonable level of automatic fire detection. This section recognizes that some jurisdictions adopting the IFC may have already implemented a high-rise retrofit program in accordance with Appendix M, and where that is the case, this section would not apply. Likewise, some jurisdictions, such as the state of Nevada, implemented high-rise retrofit programs that did not require sprinkler systems to fully comply with NFPA 13. This proposal would not apply to buildings that completed a previously approved retrofit, even if the sprinkler system is not compliant with NFPA 13.

In summary, this section should not be read as a suggestion that any high-rise building is sufficiently safe without an automatic sprinkler system. However, recognizing that the risk of catastrophic outcomes associated with high-rise fires are a function of building height and the lack of other safety features, this section advances the idea of incremental progress toward a goal as opposed to no progress at all.

1103.5.5 Pyroxylin plastics. An *automatic sprinkler system* shall be provided throughout existing buildings where cellulose nitrate film or pyroxylin plastics are manufactured, stored or handled in quantities exceeding 100 pounds (45 kg). Vaults located within buildings for the storage of raw pyroxylin shall be protected with an *approved automatic sprinkler system* capable of discharging 1.66 gallons per minute per square foot (68 L/min/m²) over the area of the vault.

C This section requires an approved automatic fire-extinguishing system in existing buildings used for the manufacture and storage of pyroxylin plastics. Cellulose nitrate (pyroxylin) plastics pose unusual and substantial fire risks. Pyroxylin plastics are the most dangerous and unstable of all plastic compounds. The chemically bound oxygen in their structure permits them to burn vigorously in the absence of atmospheric oxygen. Although these compounds produce approximately the same amount of energy as paper when they burn, pyroxylin plastics burn at a rate as much as 15 times greater than comparable common combustibles. When burning, these materials release highly flammable and toxic combustion byproducts. Consequently, cellulose nitrate fires are very difficult to control. Although this section specifies a sprinkler threshold quantity of 100 pounds (45 kg), the need for additional fire protection should be considered for pyroxylin plastics in any amount.

Although the code includes cellulose nitrate “film” in its requirements, cellulose nitrate motion picture film has not been used in the United States since the 1950s. All motion picture film produced since that time is what is typically called “safety film.” Consequently, the only application for this section relative to motion picture film is where it may be used in laboratories or storage vaults that are dedicated to film restoration and archives. The protection of these facilities is addressed in Sections 306.2 and 6504.2.

Even though this section would permit an approved fire-extinguishing system in existing structures containing pyroxylin plastics, vaults must be protected by an approved automatic sprinkler system with a density of 1.66 gpm/ft² (68 L/min/m²) over the entire area of the vault. The high sprinkler density recognizes the need to immerse the pyroxylin plastics in water to counteract the vigorous burn rate of these materials.

Note that in Table 1103.1, on the row for Section 1103.5.5, an “R” appears for all occupancy groups since pyroxylin plastics may be found in any occupancy.

1103.6 Standpipes. Existing structures shall be equipped with standpipes installed in accordance with Section 905 where required in Sections 1103.6.1 and 1103.6.2. The *fire code official* is authorized to approve the installation of manual standpipe systems to achieve compliance with this section where the responding fire department is capable of providing the required hose flow at the highest standpipe outlet.

C This section recognizes that some existing buildings do not have a standpipe system and introduces the minimum requirements for existing buildings. In place of a standpipe system that has a water supply capable of supplying the system demand automatically, this section would permit the fire code official to approve the use of a manual standpipe system in an existing building if adequate hose flow is available when the fire department charges the system. The standpipe system may be either a manual dry or a manual wet system.

1103.6.1 Existing multiple-story buildings. Existing buildings with occupied floors located more than 50 feet (15 240 mm) above the lowest level of fire department access or more than 50 feet (15 240 mm) below the highest level of fire department access shall be equipped with standpipes.

C Although it would be inappropriate to require a standpipe system in an existing building to be designed based on the requirements of Section 905.3 for new construction, this section establishes a maximum height limitation for existing buildings regardless of occupancy beyond which a standpipe is required. The system is to be installed in accordance with Section 905. Many existing buildings may also not be equipped with an automatic sprinkler system. This section will, at least, provide some means of manual fire suppression in buildings where exterior fire department access is limited.

1103.6.2 Existing helistops and heliports. Existing buildings with a rooftop helistop or heliport located more than 30 feet (9144 mm) above the lowest level of fire department access to the roof level on which the helistop or heliport is located shall be equipped with standpipes in accordance with Section 2007.5.

C A heliport or helistop on a new or existing building is a distinct hazard that can involve flammable fuels and combustible cargo and present an immediate rescue situation. In the event of an emergency, rapid deployment of hand hose lines will be necessary to attack a resulting fire, effectuate rescue and to protect exposures and the remainder of the building. This section specifies that when existing buildings have a helistop or heliport on a rooftop location more than 30 feet (9144 mm) above the lowest level of fire department vehicle access, a standpipe system is required throughout the building, not just a hose connection at the roof level. This is critical in firefighting operations because, many times, the connection below the rooftop level may be needed to gain access to the roof. If the only connection is on the roof, it is of no use if the firefighters cannot get to it. Additionally, a heliport includes fueling operations. It is entirely possible for a spill to not only affect the rooftop, but also floors below as the liquid fuel spreads. The standpipe system will again be utilized in these situations.

Note that this section refers to Section 2007.5 for design and standpipe hose connection location for the rooftop heliport or helistop. See the commentary to Sections 905.3.5 and 2007.5 for further information.

1103.7 Fire alarm systems. An *approved fire alarm system* shall be installed in existing buildings and structures in accordance with Sections 1103.7.1 through 1103.7.6 and provide occupant notification in accordance with Section 907.5 unless other requirements are provided by other sections of this code.

Exception: Occupancies with an existing, previously *approved fire alarm system*.

C This section specifies the occupancy conditions where an approved fire alarm system is retroactively required in an existing building. These systems must, upon activation, provide occupant notification throughout the area protected by the system unless other alternative provisions, such as an emergency voice/alarm communication system, are allowed.

The exception recognizes the infeasibility of requiring existing, previously approved fire alarm systems to conform to current code requirements. The existing fire alarm system must be adequately tested and maintained in accordance with Section 901.6 and shown not to create a hazard.

A common concern in the application of the exception is in the scoping of the condition. Where the building is in use and continues in use as it was before, the exception is quite clear that no modifications are required if the existing system has been previously approved, regardless of how long ago that may have been. If there is an addition to the building, the addition must meet the requirements for new work and tie into the existing system if the existing system is capable of adequately handling the new devices. The addition is only adding devices and does not require the existing system to be upgraded.

If the building is altered, the alarm system must be altered in accordance with its original installation. For example, if the existing system does not include visual notification devices and the alteration includes remodeling to existing toilet rooms, there is no requirement for installing visual devices in the remodeled toilet rooms. If, on the other hand, the existing system is not capable of receiving new devices or relocated existing devices, then the replacement of the existing panel must be considered since the altered system would be less effective than before the alteration.

1103.7.1 Group E. A *fire alarm system* shall be installed in existing Group E occupancies in accordance with Section 907.2.3.

Exceptions:

1. A manual *fire alarm system* is not required in a building with a maximum area of 1,000 square feet (93 m²) that contains a single classroom and is located not closer than 50 feet (15 240 mm) from another building.
2. A manual *fire alarm system* is not required in Group E occupancies with an *occupant load* less than 50.

C Group E occupancies are limited to educational purposes through the 12th grade. Because of the typical age and maturity of the occupants, more time may be needed to safely evacuate the building. The requirement for retroactive installation of, at least, a fire alarm system recognizes that many existing, previously approved nonsprinklered educational facilities would most likely require sprinklers under current code provisions (see Section 903.2.3).

Although not limited to this use condition, Exception 1 recognizes the current use of mobile trailer-type facilities on site as additional educational classroom facilities. This exception does not exempt the main building, but it would exempt these auxiliary buildings of limited size that do not present an exposure hazard because of the required separation distance.

Exception 2 would exempt small day care centers that serve children older than 2 years of age, a small Sunday school classroom at a church or similar limited educational use areas (see commentary, Section 907.2.3).

1103.7.2 Group I-1. An automatic *fire alarm system* shall be installed in existing Group I-1 facilities in accordance with Section 907.2.6.1.

Exception: Where each sleeping room has a *means of egress* door opening directly to an exterior egress balcony that leads directly to the *exits* in accordance with Section 1021, and the building is not more than three stories in height.

C Group I-1 facilities are assumed to have more than 16 occupants who, because of their age, mental disability or other reasons, must live in a supervised environment 24 hours a day. This section would require existing Group I-1 occupancies to have an approved fire alarm system as required by Section 907.2.6.1.

The exception recognizes the increased degree of life safety resulting from having direct access to the exterior from the sleeping rooms. The occupants are not forced to evacuate through the interior of the building during a potential fire. The exterior egress balconies must be sufficiently open to the atmosphere and constructed to minimize the accumulation of smoke and toxic gases.

1103.7.3 Group I-2. In Group I-2, an automatic *fire alarm system* shall be installed in accordance with Section 1105.10.

C This section is simply a pointer to the specific requirements for Group I-2 occupancies found in Section 1105. Section 1105.10 addresses the need for a fire alarm system. See the commentary to Section 1105.10.

1103.7.4 Group I-3. An automatic and manual *fire alarm system* shall be installed in existing Group I-3 occupancies in accordance with Section 907.2.6.3.

C Because occupants may be restrained, leading to difficulties in evacuating, an approved fire alarm system is required in existing Group I-3 occupancies. The system must comply with Section 907.2.6.3 (see commentary, Section 907.2.6.3).

1103.7.5 Group R-1. A *fire alarm system* and smoke alarms shall be installed in existing Group R-1 occupancies in accordance with Sections 1103.7.5.1 through 1103.7.5.2.1.

C This section introduces the retroactive fire alarm system and smoke alarm requirements for Group R-1 transient occupancies such as hotels, motels, and boarding houses.

1103.7.5.1 Group R-1 hotel and motel manual fire alarm system. A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in existing Group R-1 hotels and motels more than one story in height or with more than 20 *dwelling units* or *sleeping units* in aggregate.

Exceptions:

1. A manual *fire alarm system* is not required in buildings less than two stories in height where all *dwelling units*, *sleeping units*, attics and crawl spaces are separated by 1-hour *fire-resistance-rated* construction and each *sleeping unit* has direct access to a *public way*, *egress court* or yard.
2. A manual *fire alarm system* is not required in buildings not more than three stories in height with not more than 20 *dwelling units* or *sleeping units* in aggregate and equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2.
3. Manual fire alarm boxes are not required throughout the building where the following conditions are met:
 - 3.1. The building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2.
 - 3.2. The notification appliances will activate upon sprinkler water flow.
 - 3.3. Not less than one manual fire alarm box is installed at an *approved* location.

C This section specifies the conditions where a fire alarm system is required in existing Group R-1 hotels and motels. The two main criteria are independent of each other in that a fire alarm system is required if the building is more than one story in height, regardless of the number of dwelling units or sleeping units, or contains 20 dwelling units or sleeping units, regardless of the number of stories. Occupants of these types of Group R-1 facilities are assumed to be more transient than the occupants of Group R-1 boarding and rooming houses regulated by Section 907.2.8. Fires in some older multiple-story non-sprinklered hotels without fire alarm systems, as was allowed in the IBC/IFC up to 2003, have resulted in multiple fire fatalities and fire injuries, where a fire on the first floor envelopes the second story means of egress. Note that the section clarifies it is an aggregate of dwelling and sleeping units. For example, if there are 15 sleeping units and 10 dwelling units a fire alarm system would be required.

Such fires in two-story motels without automatic sprinkler systems occurred on January 4, 2010, in South Birmingham, AL (four fatalities on the second floor); on December 14, 2013, in Wausau, WI (20 injuries); and on March 21, 2014, in Point Pleasant Beach, NJ (four fatalities on the second floor).

One additional example of a multiple-life-loss fire was the Newport, OR, City Center Motel on August 5, 2016: four civilian fire deaths and three civilian fire injuries. Fire deaths occurred on both the ground floor and the second floor. The fire started on the ground floor. No sprinklers and no reports of a fire alarm system sounding.

These fires resulted in fatalities, as second floor occupants could not escape due to the means of egress being blocked by smoke and flames. In these fires resulting in fatalities, no fire alarm system was in place to provide early warning to occupants.

Exception 1 allows the omission of a fire alarm system as it recognizes the increased level of life safety afforded by adequate compartmentation using 1-hour fire-resistance-rated construction between dwelling units, sleeping units and direct exterior access for egress. Sleeping units would still require smoke alarm coverage to provide an alarm for each unit.

Exception 2 recognizes the use of a sprinkler system in place of a manual fire alarm system under two conditions. First, the building is limited to three stories and the number of dwelling units or sleeping units in aggregate remains limited to 20.

Exception 3 does not omit fire alarm systems, but rather permits them to be initiated automatically by sprinkler system water flow switches in lieu of manual fire alarm boxes. The sprinkler system is to be equipped with local audible alarms that can be heard throughout the building and at least one manual fire alarm box installed at an approved location.

The exceptions do not affect the independent provision in Section 1103.8 for single- or multiple-station smoke alarms.

1103.7.5.1.1 Group R-1 hotel and motel automatic smoke detection system. An *automatic smoke detection system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in existing Group R-1 hotels and motels throughout all interior *corridors* serving sleeping rooms not equipped with an *approved*, supervised *automatic sprinkler system* installed in accordance with Section 903.

Exception: An *automatic smoke detection system* is not required in buildings that do not have interior *corridors* serving *dwelling units* or *sleeping units* and where each *dwelling unit* or *sleeping unit* has a *means of egress* door opening directly to an *exit* or to an exterior *exit access* that leads directly to an *exit*.

C This section requires an automatic smoke detection system within interior corridors serving nonsprinklered sleeping rooms. Such systems make use of smoke detectors for alarm initiation, in accordance with Section 907.4.3, with one exception.

The exception provides that automatic smoke detection is not required in motels and hotels that do not have interior corridors and in which dwelling units or sleeping units have a door opening directly to an exterior exit access that leads directly to the exits. The exception is intended to ensure that the exit access from the sleeping unit door be exterior and not require reentering the building prior to entering the exit. Since the exit access is outside, the need for detectors other than in sleeping units is greatly reduced. Unlike Exception 1 to Section 1103.7.5.1, exit balconies and exterior stairs serving sleeping units are allowed. This section applies to those hotels and motels that may have a mix of “suites” with cooking facilities of some type and sleeping units that have only beds and sanitation.

1103.7.5.2 Group R-1 boarding and rooming houses manual fire alarm system. A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in existing Group R-1 boarding and rooming houses.

Exception: Buildings less than two stories in height where all *sleeping units*, attics and crawl spaces are separated by 1-hour *fire-resistance-rated* construction and each *dwelling unit* or *sleeping unit* has direct access to a *public way*, *egress court* or *yard*.

C Group R-1 boarding and rooming houses are still assumed to be transient residential occupancies even though the functional use of a boarding and rooming house is different from that of a typical hotel or motel. Boarding and rooming houses tend to have more extended living arrangements and border on being classified as Group R-2 facilities. For this reason, this section requires Group R-1 boarding and rooming houses to be equipped with a manual fire alarm system regardless of the height of the building or number of sleeping rooms.

The exception recognizes the increased level of life safety afforded by adequate compartmentation using 1-hour fire-resistance-rated construction between dwelling units and sleeping units and direct exterior access for egress.

1103.7.5.2.1 Group R-1 boarding and rooming houses automatic smoke detection system. An *automatic smoke detection system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in existing Group R-1 boarding and rooming houses throughout all interior *corridors* serving *dwelling units* or *sleeping units* not equipped with an *approved*, supervised sprinkler system installed in accordance with Section 903.

Exception: Buildings equipped with single-station smoke alarms meeting or exceeding the requirements of Section 907.2.11.1 and where the *fire alarm system* includes not less than one manual fire alarm box per floor arranged to initiate the alarm.

C This section requires an automatic smoke detection system within interior corridors serving dwelling units and sleeping units. Such systems make use of smoke detectors for alarm initiation, in accordance with Section 907.4.3, with one exception.

The exception provides that automatic fire detectors are not required in boarding and rooming houses where single-station smoke alarms complying with the minimum requirements of Section 907.2.11.1 and at least one manual fire alarm box per floor are installed. The single-station smoke alarms give the desired early warning notification to occupants and the manual fire alarm box is an additional means to activate the occupant notification system.

1103.7.6 Group R-2. A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in existing Group R-2 occupancies more than three stories in height or with more than 16 *dwelling* or *sleeping units*.

Exceptions:

1. Where each living unit is separated from other contiguous living units by *fire barriers* having a *fire-resistance rating* of not less than $\frac{3}{4}$ hour, and where each living unit has either its own independent *exit* or its own independent *stairway* or *ramp* discharging at grade.
2. A separate *fire alarm system* is not required in buildings that are equipped throughout with an *approved* supervised *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2 and having a local alarm to notify all occupants.
3. A *fire alarm system* is not required in buildings that do not have interior *corridors* serving *dwelling units* and are protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2, provided that *dwelling units* either have a *means of egress* door opening directly to an exterior *exit access* that leads directly to the *exits* or are served by open-ended *corridors* designed in accordance with Section 1027.6, Exception 3.

4. A *fire alarm system* is not required in buildings that do not have interior *corridors* serving *dwelling units*, do not exceed three stories in height and comply with both of the following:
 - 4.1. Each *dwelling unit* is separated from other contiguous *dwelling units* by *fire barriers* having a *fire-resistance rating* of not less than $\frac{3}{4}$ hour.
 - 4.2. Each *dwelling unit* is provided with smoke alarms complying with the requirements of Section 907.2.11.

C This section specifies the conditions where a manual fire alarm system is required in existing Group R-2 apartment buildings based on height or the number of dwelling or sleeping units. Occupants of Group R-2 facilities tend to be more permanent than those in Group R-1 facilities.

Exception 1 recognizes the increased degree of life safety afforded by compartmentation using fire-resistance-rated construction and independent means of egress (see Section 907.2.9).

As indicated in Exception 2, existing buildings that are equipped throughout with an automatic sprinkler system in accordance with NFPA 13 or 13R do not need a manual fire alarm system if local alarms will sound upon activation of the sprinkler system. The exception essentially eliminates the need for manual fire alarm boxes if evacuation alarms can still be heard throughout the building upon sprinkler system water flow. Other alarm-initiating devices would similarly not be required if the water flow device will initiate the appropriate alarm. This exception does not affect the independent provisions in Section 907.2.11.

Exception 3 mirrors Section 907.2.9.1, Exception 3, and recognizes the superior fire record of sprinklered multiple-family occupancies by allowing omission of a fire alarm system where a building is fully sprinklered, has no interior egress corridors and provides direct exterior egress from each dwelling unit. Note that in such buildings, rated fire separations are still required between units.

Exception 4 provides a reasonable alternative to retrofitting a manual fire alarm system in existing Group R-2 occupancy buildings not exceeding three stories in height and having exits that lead directly to the outside. Fire risk in apartments tends to be greatest for occupants inside the dwelling unit where a fire originates, and money spent to retrofit fire safety equipment in apartments is better spent within dwelling units, as opposed to common areas.

Countless existing apartment buildings have only a single smoke alarm in the common area. The code does not require retrofitting sleeping rooms with smoke alarms where such alarms weren't required at the time of construction. The lack of smoke alarms in bedrooms, and particularly the lack of interconnecting alarm signals, increases the risk of injury or death in a unit of fire origin and other units that experience smoke infiltration. An additional consequence may be delayed recognition of a fire event, which increases the risk of harm to other building occupants and may delay notification of the fire department. This exception also requires minimal separation from other units.

1103.8 Single- and multiple-station smoke alarms. Single- and multiple-station smoke alarms shall be installed in existing Group I-1 and R occupancies in accordance with Sections 1103.8.1 through 1103.8.3.

C This section introduces the requirements for the installation of smoke alarms in existing Group I-1 and R occupancies. These requirements recognize the benefit of installing smoke alarms in existing structures but provide several exceptions for buildings that are not undergoing substantial renovations. These provisions also correlate with Section 704.6 of the *International Property Maintenance Code*® (IPMC®) for single- or multiple-station smoke alarm requirements for existing Group R dwellings.

1103.8.1 Where required. Existing Group I-1 and R occupancies shall be provided with single-station smoke alarms in accordance with Section 907.2.11. Interconnection and power sources shall be in accordance with Sections 1103.8.2 and 1103.8.3, respectively.

Exceptions:

1. Where the code that was in effect at the time of construction required smoke alarms and smoke alarms complying with those requirements are already provided.
2. Where smoke alarms have been installed in occupancies and *dwellings* that were not required to have them at the time of construction, additional smoke alarms shall not be required provided that the existing smoke alarms comply with requirements that were in effect at the time of installation.
3. Where smoke detectors connected to a *fire alarm system* have been installed as a substitute for smoke alarms.

C This section requires that Group I-1 and R occupancies be provided with single-station smoke alarms where they are not already provided. Reference is made to the new building requirements of Section 907.2.11 for the primary requirements; however, interconnection and power supply are more specifically addressed for existing buildings in Sections 1103.8.2 and 1103.8.3. Therefore, the basic requirement is that smoke alarms must be provided in all locations required by Section 907.2.11, but several modified requirements related to interconnection of alarms and power supply are provided, recognizing the impracticality of such installations in existing conditions. Three exceptions are also provided to address possible scenarios where smoke alarms have already been installed but the installation does not meet the current code requirements, recognizing that the code is intended to permit existing smoke alarm installations to continue unchanged where they meet the code that was in effect at the time they were installed.

Exception 1 indicates that smoke alarms installed and maintained in accordance with the applicable code at the time of construction can continue unchanged. Exception 2 indicates that smoke alarms that were not required by the code at the time of construction but were later installed can continue where they meet the requirements of the applicable code at the time of installation. Exception 3 indicates that smoke detectors connected to a fire alarm system may be used in lieu of smoke alarms.

In summary, this section requires the installation of smoke alarms in Group I-1 and R occupancies that do not currently have any smoke alarms. It does not require compliance with the current smoke alarm requirements if the building already has smoke alarms meeting requirements that were applicable when they were installed. The focus here is not to have the owner replace or revise his or her smoke alarms any time the code requirements for new construction change.

1103.8.2 Interconnection. Where more than one smoke alarm is required to be installed within an individual *dwelling or sleeping unit*, the smoke alarms shall be interconnected in such a manner that the activation of one alarm will activate all of the alarms in the individual unit. Physical interconnection of smoke alarms shall not be required where *listed* wireless alarms are installed and all alarms sound upon activation of one alarm. The alarm shall be clearly audible in all bedrooms over background noise levels with all intervening doors closed.

Exceptions:

1. Interconnection is not required in buildings that are not undergoing *alterations*, repairs or construction of any kind.
2. Smoke alarms in existing areas are not required to be interconnected where *alterations* or repairs do not result in the removal of interior wall or ceiling finishes exposing the structure, unless there is an attic, crawl space or *basement* available that could provide access for interconnection without the removal of interior finishes.

C This section, like Section 907.2.11.5, requires that where multiple-station smoke alarms are present, they are to be interconnected and be audible over back-ground noises. This section also includes language that would allow listed wireless alarms to substitute for wired interconnection of the smoke alarms in both new and existing construction, making it clear that listed wireless systems comply with the code. Historically, while some code officials have not recognized wireless interconnection of smoke alarms as meeting the code requirement for interconnection, most have accepted such interconnection as its use has proliferated and its reliability has been demonstrated. See the commentary to Section 907.2.11.5 for further information on interconnection.

Exception 1 does not require interconnection where the building is not undergoing any construction or repairs. Exception 2 clarifies to what extent the building must be undergoing construction before interconnection is required. Generally, the exceptions try to be reasonable based on the practicality of such installations; therefore, unless areas such as attics or crawl spaces can still be utilized while the interior finishes are being removed (e.g., drywall removed, exposing the studs), interconnection would not be required. Such renovations may only be limited to portions of a structure; therefore, complete interconnection may not be practical or possible. The intent is that additional walls, etc., should not be removed solely to interconnect the smoke alarms. Battery-powered alarms, as allowed by Section 1103.8.3, may be interconnected where a building is not supplied by a commercial power source.

1103.8.3 Power source. Single-station smoke alarms shall receive their primary power from the building wiring provided that such wiring is served from a commercial source and shall be equipped with a battery backup. Smoke alarms with integral strobes that are not equipped with battery backup shall be connected to an emergency electrical system. Smoke alarms shall emit a signal when the batteries are low. Wiring shall be permanent and without a disconnecting switch other than as required for overcurrent protection.

Exceptions:

1. Smoke alarms are permitted to be solely battery operated in existing buildings where construction is not taking place.
2. Smoke alarms are permitted to be solely battery operated in buildings that are not served from a commercial power source.
3. Smoke alarms are permitted to be solely battery operated in existing areas of buildings undergoing *alterations* or repairs that do not result in the removal of interior walls or ceiling finishes exposing the structure, unless there is an attic, crawl space or *basement* available that could provide access for building wiring without the removal of interior finishes.

C This section requires both a primary power supply from the building wiring, as well as a supervised battery backup. Wiring must be permanent and supplied from a circuit that is properly overcurrent-protected, but without any on-off switching capability. Currently, there are some smoke alarms on the market that have an integral visual alarm notification component (a strobe light) but do not have a built-in battery for the strobe. Thus, if the power for the building fails, the smoke detection and audible signal of the device may still operate, but the strobe will not. It is critical for rooms that may house those with hearing impairments and are equipped with these smoke alarms that the strobe be as reliable as the audible signal element. This section resolves that concern by requiring such smoke alarms to be connected to an emergency electrical system for backup power in accordance with Section 1203.

This section is very similar to Section 907.2.11. The primary differences are the exceptions.

Exception 1 allows the use of batteries as the sole power source if no construction or related repairs are occurring in the building.

Exception 2 allows the use of batteries as the sole power source if there is no other power source available (i.e., the building is without an electric utility connection).

Exception 3 clarifies to what extent the building must be undergoing construction before connection to the building wiring is required. Generally, the exceptions try to be reasonable based on the practicality of such installations; therefore, unless areas such as attics or crawl spaces can be utilized for interstitial access for wiring while the interior finishes are being removed (e.g., drywall removed, exposing the studs), connection to the building wiring would not be required. Such renovations may only be limited to portions of a structure; therefore, adding permanent wiring may not be practical or possible. The intent is that additional walls, etc., should not be removed solely to connect the smoke alarms to the building wiring.

1103.9 Carbon monoxide detection. Carbon monoxide detection shall be installed in existing buildings where any of the conditions identified in Section 915.1.1 exist. Carbon monoxide alarms shall be installed in the locations specified in Section 915.2 and the installation shall be in accordance with Section 915.4.

Exceptions:

1. Carbon monoxide alarms are permitted to be solely battery operated where the code that was in effect at the time of construction did not require carbon monoxide detectors to be provided.
2. Carbon monoxide alarms are permitted to be solely battery operated in *dwelling units* that are not served from a commercial power source.
3. A carbon monoxide detection system in accordance with Section 915.5 shall be an acceptable alternative to carbon monoxide alarms.

C This section requires CO detection in accordance with the requirements of Section 915. Section 915 requires that CO detection be provided for all buildings where carbon monoxide exposure occurs. The specific requirements for alarm and detector placement are addressed in Section 915. This recognizes the need for and the practicality of providing such protection in existing buildings. Use of the term “detection” includes both carbon monoxide alarms and detection systems. The key is that occupancies have the ability to detect CO.

Exceptions 1 and 2 acknowledge that allowing battery-powered units to be provided is a relatively low-cost solution in existing buildings where carbon monoxide alarms were not initially required or where dwelling units and sleeping units are not connected to a commercial power source. Exception 3 provides the option of using a CO detection system in place of individual CO alarms. It is important to also understand that CO alarms have specific listings. In addition, placement of a CO alarm in a location distant from those you are trying to protect may be of little use as they will not be able to hear the alarm. See the commentary to Section 915 for more details on alarm requirements and placement.

1103.10 Medical gases. Medical gases stored and transferred in health-care-related facilities shall be in accordance with Chapter 53.

C This section requires compliance with Chapter 53 compressed gases where medical gases are stored and transferred in existing health-care-related facilities. This will address several different types of occupancies including Group I-2 and ambulatory care facilities but can also address facilities such as a small dentist office. It was considered necessary to make sure that all existing facilities comply with these requirements to meet CMS guidelines. A general reference was made since the medical gas requirements of Section 5306 are only one aspect of the regulation of compressed gases.

SECTION 1104—MEANS OF EGRESS FOR EXISTING BUILDINGS

1104.1 General. *Means of egress* in existing buildings shall comply with the minimum egress requirements where specified in Table 1103.1 as further enumerated in Sections 1104.2 through 1104.25, and the building code that applied at the time of construction. Where the provisions of this chapter conflict with the building code that applied at the time of construction, the most restrictive provision shall apply. Existing buildings that were not required to comply with a building code at the time of construction shall comply with the minimum egress requirements where specified in Table 1103.1 as further enumerated in Sections 1104.2 through 1104.25.

C The primary concept of this section is to require existing buildings to comply with the specific means of egress requirements for new buildings except as modified by this section. Where an item is specifically addressed by Section 1104, the requirements of this section allow existing buildings to not fully meet the requirements for new buildings in Sections 1003 through 1031.

For example, the guard height requirements in Section 1104.6 include exceptions for the guard height requirement for new buildings in Section 1015.2. The requirements for existing buildings in Section 1104 are intended to be the same or less stringent than the comparable provisions for new buildings in Sections 1003 through 1031.

This section applies to most existing buildings. Though the code does not specifically state, where a building has been built to meet the requirements of the building code in effect at the time of construction and, in the opinion of the fire code official, the means of egress is not hazardous, the building would satisfy this section. As such, none of the specific requirements in this section would apply.

In the absence of any evidence of compliance with a building code at the time the building was constructed, all provisions of Sections 1104.1 through 1104.25 apply. See Chapter 13 of the IEBC for a performance-based compliance alternative analysis regimen for existing buildings, especially Sections 1305.2.11 through 1305.2.15, which deal with the means of egress analysis. Section 601 of the *International Code Council Performance Code*® (ICCPC®) also provides a performance approach to means of egress evaluation.

1104.2 Elevators, escalators and moving walks. Elevators, escalators and moving walks shall not be used as a component of a required *means of egress*.

Exceptions:

1. Elevators used as an *accessible means of egress* where allowed by Section 1009.4.
2. Previously *approved* elevators, escalators and moving walks in existing buildings.

C This section is the same as Section 1003.7, except Exception 2 is added. Thus, an escalator or moving walk could be used as part of the required means of egress in an existing building if it had been previously approved by the fire code official.

1104.3 Exit sign illumination. Exit signs shall be internally or externally illuminated. The face of an exit sign illuminated from an external source shall have an intensity of not less than 5 footcandles (54 lux). Internally illuminated signs shall provide equivalent luminance and be *listed* for the purpose.

Exception: *Approved* self-luminous signs that provide evenly illuminated letters shall have a minimum luminance of 0.06 foot-lamberts (0.21 cd/m²).

C This section is the same as Section 1013.3 for new buildings, except that Section 1013.3 includes an exception to illumination for tactile signs. The same exception should apply for existing buildings.

1104.4 Power source. Where emergency illumination is required in Section 1104.5, exit signs shall be visible under emergency illumination conditions.

Exception: *Approved* signs that provide continuous illumination independent of external power sources are not required to be connected to an emergency electrical system.

C This section requires that exit signs serving the occupancies listed in Section 1104.5 be illuminated during the use of emergency power for the means of egress. The comparable section for new buildings is Section 1013.6.3. Exit signs for all new building occupancies must be illuminated at all times and have an emergency power source.

1104.5 Illumination emergency power. Where *means of egress* illumination is provided, the power supply for *means of egress* illumination shall normally be provided by the premises' electrical supply. In the event of power supply failure, illumination shall be automatically provided from an emergency system for the following occupancies where such occupancies require two or more *means of egress*:

1. Group A having 50 or more occupants.

Exception: Assembly occupancies used exclusively as a place of worship and having an *occupant load* of less than 300.

2. Group B buildings three or more stories in height, buildings with 100 or more occupants above or below a *level of exit discharge* serving the occupants or buildings with 1,000 or more total occupants.
3. Group E in interior *exit access* and *exit stairways* and *ramps, corridors*, windowless areas with student occupancy, shops and laboratories.
4. Group F having more than 100 occupants.

Exception: Buildings used only during daylight hours and that are provided with windows for natural light in accordance with the *International Building Code*.

5. Group I.
6. Group M.

Exception: Buildings less than 3,000 square feet (279 m²) in gross sales area on one story only, excluding mezzanines.

7. Groups R-1 and R-2.

Exception: Where each *dwelling unit* or *sleeping unit* has direct access to the outside of the building at grade.

C Section 1008.3 requires emergency power for emergency egress lighting in all new buildings along aisles, corridors and exit access stairways leading through or from spaces where two or more means of egress are required. This section requires emergency power for emergency egress lighting only for the listed occupancy conditions for existing buildings.

1104.5.1 Emergency power duration and installation. Emergency power for *means of egress* illumination shall be provided in accordance with Section 1203. In other than Group I-2, emergency power shall be provided for not less than 60 minutes for systems requiring emergency power.

C A 90-minute emergency power duration for emergency egress lighting is required by Section 1008.3 for new buildings. The emergency power system for emergency egress lighting in existing buildings in other than Group I-2 occupancies, however, is to provide power for at least 60 minutes. This duration is consistent with Section 1104.1.

1104.6 Guards. Guards complying with this section shall be provided at the open sides of *means of egress* that are more than 30 inches (762 mm) above the floor or grade below.

C Where guards are required in this section, the requirements are the same as in Section 1015 for new buildings (see commentary, Section 1015.1). Sections 1104.6.1 and 1104.6.2 do provide allowances for the height and opening limitations that are less than required for new construction.

1104.6.1 Height of guards. Guards shall form a protective barrier not less than 42 inches (1067 mm) high.

Exceptions:

1. Existing guards on the open side of *exit access* and *exit stairways* and *ramps* shall be not less than 30 inches (760 mm) high.
2. Existing *guards* within *dwelling units* shall be not less than 36 inches (910 mm) high.
3. Existing *guards* in assembly seating areas.

C Guard height in new construction is typically 42 inches (1067 mm) minimum. In the legacy codes, there were allowances for guards to have the top rail at the height required for handrails where the guard was along a stairway or ramp (except for Group E). Exception 1 is in recognition of those allowances. Exceptions 2 and 3 are in recognition of the lower guard heights allowed within dwelling units (Section 1015.3, Exceptions 1, 2, 3 and 4) and with assembly seating (Section 1015.3, Exception 5 and Section 1030.17) for new construction.

1104.6.2 Opening limitations. Open *guards* shall have balusters or ornamental patterns such that a 6-inch-diameter (152 mm) sphere cannot pass through any opening up to a height of 34 inches (864 mm).

Exceptions:

1. At elevated walking surfaces for access to, and use of, electrical, mechanical or plumbing systems or equipment, *guards* shall have balusters or be of solid materials such that a sphere with a diameter of 21 inches (533 mm) cannot pass through any opening.
2. In occupancies in Group I-3, F, H or S, the clear distance between intermediate rails measured at right angles to the rails shall not exceed 21 inches (533 mm).
3. *Approved* existing open *guards*.

C Generally, this section allows maximum openings of 6 inches (152 mm) versus the maximum openings of 4 inches (102 mm) allowed by Section 1015.4. Exception 1 mirrors Section 1015.4, Exception 3, and Exception 2 mirrors Section 1015.4, Exception 4.

1104.7 Size of doors. The required capacity of each door opening shall be sufficient for the *occupant load* thereof and shall provide a minimum clear opening width of 28 inches (711 mm). Where this section requires a minimum clear opening width of 28 inches (711 mm) and a door opening includes two door leaves without a mullion, one leaf shall provide a clear opening width of 28 inches (711 mm). The minimum clear opening height of doorways shall be 80 inches (2032 mm).

Exceptions:

1. The minimum and maximum width shall not apply to door openings that are not part of the required *means of egress* in occupancies in Group R-2 and R-3 units that are not required to be an Accessible unit, Type A unit or Type B unit.
2. Door openings to storage closets less than 10 square feet (0.93 m²) in area shall not be limited by the minimum clear opening width.
3. The width of door leaves in revolving doors that comply with Section 1010.3.1 shall not be limited.
4. The maximum width of door leaves in power-operated doors that comply with Section 1010.3.2 shall not be limited.
5. Door openings within a *dwelling unit* shall have a minimum clear opening height of 78 inches (1981 mm).
6. In *dwelling* and *sleeping units* that are not required to be Accessible units, Type A units or Type B units, exterior door openings, other than the required exit door, shall have a minimum clear opening height of 76 inches (1930 mm).
7. *Exit access* doors serving a room not larger than 70 square feet (6.5 m²) shall have a minimum door leaf width of 24 inches (610 mm).
8. The minimum clear opening width shall not apply to doors for nonaccessible showers or sauna compartments.
9. The minimum clear opening width shall not apply to the doors for nonaccessible toilet stalls.
10. Door closers, overhead door stops, frame stops, power door operators and electromagnetic door locks shall be permitted to project into the door opening height not lower than 78 inches (1980 mm) above the floor.

C Generally, this section requires doors to provide a minimum clear opening of 28 inches (711 mm) versus the minimum clear opening of 32 inches (813 mm) for new construction in Section 1010.1.1. The other provisions and exceptions in this section are the same as those in Section 1010.1.1 for new construction. Exceptions 2, 6 and 7 of Section 1010.1.1 are not included in this section. Exception 2 is not needed since these provisions for existing buildings only require a 28-inch clear opening (711 mm) door. Exceptions 6 and 7 relate to accessibility requirements that are not intended to be retroactively applied except within the context of the IEBC.

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1104.7.1 Group I-2. In Group I-2 occupancies, *means of egress* doors where used for the movement of beds shall provide a minimum clear opening width of 41½ inches (1054 mm).

Doors serving as *means of egress* doors and not used for movement of beds shall provide a minimum clear opening width of 32 inches (813 mm).

C In Group I-2 occupancies, the need to move patients in beds requires a clear opening of 41 1/2 inches, consistent with new construction requirements. As with ambulatory care occupancies in Section 1104.7.2, other doors within Group I-2 occupancies could use the 28-inch clear opening allowance.

1104.7.2 Ambulatory care. In ambulatory care facilities, doors serving as *means of egress* from patient treatment rooms shall provide a minimum clear opening width of 32 inches (813 mm).

C The 32-inch clear width for patient treatment or sleeping room doors within ambulatory care facilities is based on the nature of the activities within the space. As with Group I-2 occupancies in Section 1104.7.1, other doors within ambulatory care facilities could use the 28-inch clear opening allowance.

1104.8 Forces to unlatch and open doors. Forces shall be applied to the latch side. Forces required to unlatch and open doors shall be in accordance with Sections 1104.8.1 and 1104.8.2.

C This section is almost identical to Section 1010.1.3. The subsections break down the requirements by the forces required for unlatching and the force required to open doors.

1104.8.1 Unlatching doors. The forces to unlatch doors shall comply with the following:

1. Where door hardware operates by push or pull, the operational force to unlatch the door shall not exceed 15 pounds (67 N).
2. Where door hardware operates by rotation, the operational force to unlatch the door shall not exceed 28 inch-pounds (3.164 N-m).

C This section provides the specific criteria related to the maximum unlatching forces for doors. This is broken down by the type of door hardware used. These criteria are consistent with new construction.

1104.8.2 Opening doors. The forces to open doors shall comply with the following:

1. For interior swinging egress doors that are manually operated, other than doors required to be fire rated, the force for pushing or pulling open the door shall not exceed 5 pounds (22 N).
2. For other swinging doors, sliding doors or folding doors and doors required to be fire-rated, the door shall require not more than a 30-pound (133 N) force to be set in motion and shall move to a full-open position when subjected to not more than a 15-pound (67 N) force.

C This section provides the maximum forces permitted to open a swinging door depending on the application of the door. These criteria are consistent with new construction.

1104.9 Revolving doors. Revolving doors shall comply with the following:

1. A revolving door shall not be located within 10 feet (3048 mm) of the foot or top of *stairways* or escalators. A dispersal area shall be provided between the *stairways* or escalators and the revolving doors.
2. The revolutions per minute for a revolving door shall not exceed those shown in Table 1104.9.
3. Each revolving door shall have a conforming side-hinged swinging door in the same wall as the revolving door and within 10 feet (3048 mm).

Exceptions:

1. A revolving door is permitted to be used without an adjacent swinging door for street-floor elevator lobbies provided that a *stairway*, escalator or door from other parts of the building does not discharge through the lobby and the lobby does not have any occupancy or use other than as a means of travel between elevators and a street.
2. Existing revolving doors where the number of revolving doors does not exceed the number of swinging doors within 20 feet (6096 mm).

C This section is comparable with Section 1010.3.1 for new buildings. The capability of collapsing to a width of 36 inches (914 mm) in Condition 2 of Section 1010.3.1 for new construction does not apply to existing revolving doors. Exceptions 1 and 2 are not allowed for new construction.

TABLE 1104.9—REVOLVING DOOR SPEEDS		
INSIDE DIAMETER (feet-inches)	POWER-DRIVEN-TYPE SPEED CONTROL (rpm)	MANUAL-TYPE SPEED CONTROL (rpm)
6-6	11	12
7-0	10	11
7-6	9	11
8-0	9	10
8-6	8	9
9-0	8	9
9-6	7	8
10-0	7	8
For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.		

C Tables 1010.3.1(1) and 1010.3.1(2) for new construction have a wider range of door sizes and often have a slower maximum speed allowance than this table.

1104.9.1 Egress component. A revolving door used as a component of a *means of egress* shall comply with Section 1104.9 and all of the following conditions:

1. Revolving doors shall not be given credit for more than 50 percent of the required egress capacity.
2. Each revolving door shall be credited with not more than a 50-person capacity.
3. Revolving doors shall be capable of being collapsed when a force of not more than 130 pounds (578 N) is applied within 3 inches (76 mm) of the outer edge of a wing.

C This section is similar to Section 1010.3.1.1 for new construction.

1104.10 Stair dimensions for existing stairways. Existing *stairways* in buildings shall be permitted to remain if the rise does not exceed 8¹/₄ inches (210 mm) and the run is not less than 9 inches (229 mm). Existing *stairways* can be rebuilt.

Exception: Other *stairways* approved by the fire code official.

C This section includes a much less stringent criterion for stairway treads and risers than that in Section 1011.5.2. This section also allows existing stairways to be rebuilt. The tread and riser dimensions of the rebuilt stairway have additional allowances in Section 1104.10.1.

1104.10.1 Dimensions for replacement stairways. The replacement of an existing *stairway* in a structure shall not be required to comply with the new *stairway* requirements of Section 1011 where the existing space and construction will not allow a reduction in pitch or slope.

C This section is consistent with IEBC Section 503.1. This section is intended to allow that, where an existing stairway was built with a steeper rise/run ratio than permitted in the current code, the stairway can be replaced with its original configuration. Enlarging the opening to achieve the current rise/run ratio and headroom would be considered technically infeasible. The principle is that not allowing for this option could result in stairways that were not maintained because they could not be brought up to current codes.

1104.11 Winders. Existing winders shall be allowed to remain in use if they have a minimum tread depth of 6 inches (152 mm) and a minimum tread depth of 9 inches (229 mm) at a point 12 inches (305 mm) from the narrowest edge.

C In new construction, winders are only permitted within dwelling units and along curved or spiral stairways (Sections 1011.5.3, 1011.9 and 1011.10). The tread size allowances for existing stairways are smaller than permitted for curved stairways.

1104.12 Curved stairways. Existing curved *stairways* shall be allowed to continue in use, provided that the minimum depth of tread is 10 inches (254 mm) and the smallest radius shall be not less than twice the width of the *stairway*.

C For other than certain residential occupancies, this section has fewer requirements than Section 1011.9 for new construction. The relaxed requirements of the exception to Section 1011.9 apply to Group R-3 occupancies and individual dwelling units in Group R-2 occupancies.

1104.13 Stairway handrails. *Stairways* shall have *handrails* on at least one side. *Handrails* shall be located so that all portions of the *stairway* width required for egress capacity are within 44 inches (1118 mm) of a *handrail*.

Exception: *Aisle stairs* provided with a center *handrail* are not required to have additional *handrails*.

C Handrails are required on both sides, with certain exceptions, for new construction. This section requires rails on one side. The 44-inch (1118 mm) dimension in this section is an increase from the 30-inch (762 mm) dimension in Section 1014.10 for

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new construction. Handrails for assembly occupancies in new construction are covered in Section 1030.16. See Section 1014 for handrail requirements for new construction.

1104.13.1 Height. *Handrail* height, measured above *stair* tread nosings, shall be uniform, not less than 30 inches (762 mm) and not more than 42 inches (1067 mm).

C The allowable range of handrail height installation in this section is broader than the 34 inches (834 mm) up to 38 inches (965 mm) allowed for new construction in Section 1014.2. This is in recognition of previous allowances for the handrail to be the top rail of a guard. This is still permitted for new construction in assembly seating (see Section 1030.16, Exception 2).

1104.14 Slope of ramps. *Ramp* runs utilized as part of a *means of egress* shall have a running slope not steeper than 1 unit vertical in 10 units horizontal (10 percent slope). The slope of other *ramps* shall not be steeper than 1 unit vertical in 8 units horizontal (12.5 percent slope).

C The slope of means of egress ramps for new construction in Section 1012.2 is a maximum of 1:12. The steeper ramps of 1:8 slope in assembly aisles, as permitted by Section 1030.14 for new construction, should also apply to existing construction.

1104.15 Width of ramps. Existing *ramps* are permitted to have a minimum width of 30 inches (762 mm) but not less than the width required for the number of occupants served as determined by Section 1005.1. In Group I-2, *ramps* serving as a *means of egress* and used for the movement of patients in beds shall comply with Section 1105.6.4.

C Ramps for new construction in Section 1012.5.1 are required to have a minimum clear width of 36 inches (914 mm) between handrails. This section requires a 30-inch (762 mm) minimum ramp width (see commentary, Section 1012.5.1). This section also references the more specific requirements for Group I-2 occupancies in Section 1105.6.4. The minimum width required by that section is 48 inches. See the commentary to Section 1105.6.4.

[BE] 1104.16 Fire escape stairways. Fire escape *stairways* shall comply with Sections 1104.16.1 through 1104.16.6.

C This section and the referenced sections provide detailed requirements for fire escape stairways. Fire escape stairways are allowed as a required means of egress only in an existing building, according to IEBC Sections 504 and 804.5.1.2.1. Sections 1104.16.1 through 1104.16.6 of this code are similar to Sections 504.1 through 504.5 of the IEBC.

[BE] 1104.16.1 Existing means of egress. Fire escape *stairways* shall be permitted in existing buildings but shall not constitute more than 50 percent of the required exit capacity.

C Fire escapes are limited to serving 50 percent of the required exit capacity because they have minimum usability and are not appropriate for use by persons with limited physical capability.

[BE] 1104.16.2 Opening protectives. Doors and windows within 10 feet (3048 mm) of fire escape *stairways* shall be protected with ³/₄-hour opening protectives.

Exception: Opening protectives shall not be required in buildings equipped throughout with an *approved automatic sprinkler system*.

C It is important that doors and windows in the vicinity of a fire escape be protected so that the fire escape will be usable during a fire. The exception for an automatic sprinkler system takes into account the ability of the sprinkler system to control the fire and reduce the need for opening protectives.

[BE] 1104.16.3 Dimensions. Fire escape *stairways* shall meet the minimum width, capacity, riser height and tread depth as specified in Section 1104.10.

C The riser height and tread depth requirements are the same as for existing stairways. The limits are specified in Section 1104.10; however, the minimum width and occupant capacity are not specified.

[BE] 1104.16.4 Access. Access to a fire escape *stairway* from a *corridor* shall not be through an intervening room. Access to a fire escape *stairway* shall be from a door or window meeting the criteria of Section 1005.1. Access to a fire escape *stairway* shall be directly to a balcony, landing or platform. These shall not be higher than the floor or windowsill level and not lower than 8 inches (203 mm) below the floor level or 18 inches (457 mm) below the windowsill.

C This section establishes the arrangement of the fire escape to the building so that the fire escape can be easily reached. Access is not permitted through a room because the room could pose an unacceptable hazard or be locked. The elevation limits of the exterior landing, balcony or platform are to make the fire escape easily accessible.

[BE] 1104.16.5 Materials and strength. Components of fire escape *stairways* shall be constructed of noncombustible materials. Fire escape *stairways* and balconies shall support the dead load plus a live load of not less than 100 pounds per square foot (4.78 kN/m²). Fire escape *stairways* and balconies shall be provided with a top and intermediate *handrail* on each side.

C The noncombustible construction requirement is so that the fire escape will be available for use during a fire. The loading of 100 pounds per square foot (psf) (4.78 kN/m²) anticipates that a number of persons will be using the fire escape at the same time. To verify the ongoing viability of fire escape stairways, Section 1104.16.5.1 requires tests or field surveys to validate the capacity.

[BE] 1104.16.6 Termination. The lowest balcony shall not be more than 18 feet (5486 mm) from the ground. Fire escape *stairways* shall extend to the ground or be provided with counterbalanced *stairs* reaching the ground.

Exception: For fire escape *stairways* serving 10 or fewer occupants, an *approved* fire escape ladder is allowed to serve as the termination.

C This section controls the elevation of the lowest fire escape balcony. For fire escapes that serve 10 or fewer people, a ladder from the lowest balcony to the ground is permitted. In all other cases, the lower balcony is to be served by fire escape stairways. The counterbalance keeps the fire escape stairway up and off ground level when not in use.

1104.17 Corridor construction. *Corridors* serving an *occupant load* greater than 30 and the openings therein shall provide an effective barrier to resist the movement of smoke. Transoms, louvers, doors and other openings shall be kept closed or be self-closing. In Group I-2, *corridors* in areas housing patient sleeping or care rooms shall comply with Section 1105.5.

Exceptions:

1. *Corridors* in occupancies other than in Group H, that are equipped throughout with an *approved automatic sprinkler system*.
2. *Corridors* in occupancies in Group E where each room utilized for instruction or assembly has not less than one-half of the required *means of egress* doors opening directly to the exterior of the building at ground level.
3. *Corridors* that are in accordance with the *International Building Code*.

C This section relates to Section 1020 for new construction. This section requires the corridors to be an effective barrier to resist the movement of smoke. Section 1020.2 for new construction requires that corridors have a fire-resistance rating. Generally, solid walls and doors that would resist smoke migration are the focus of this section. The walls or doors need not have a fire-resistance rating.

A pointer is provided to the requirements for Group I-2 occupancies in areas where such occupancies contain patient sleeping and care. Section 1105.5 provides specific requirements related to the separation requirements, allowable openings. See the commentary to Section 1105.5.

1104.17.1 Corridor openings. Openings in *corridor* walls shall comply with the requirements of the *International Building Code*.

Exceptions:

1. Where 20-minute fire door assemblies are required, solid wood doors not less than 1.75 inches (44 mm) thick or insulated steel doors are allowed.
2. Openings protected with fixed wire glass set in steel frames.
3. Openings covered with 0.5-inch (12.7 mm) gypsum wallboard or 0.75-inch (19.1 mm) plywood on the room side.
4. Opening protection is not required where the building is equipped throughout with an *approved automatic sprinkler system*.

C This section makes reference to the IBC requirements for corridor openings. Section 1020.2 specifies that corridors required to be fire-resistance rated are to be constructed according to IBC Section 708 for fire partitions. IBC Section 708.6 refers to IBC Section 716 for protected openings in smoke partitions. Thus, IBC Section 716 applies to existing buildings where a fire-resistance-rated corridor would be required for new construction, except where one of the exceptions in this section applies. The exceptions in this section provide a number of practical alternatives for existing buildings.

1104.18 Dead ends. Where more than one *exit* or *exit access doorway* is required, the *exit access* shall be arranged such that dead ends do not exceed the limits specified in Table 1104.18.

Exceptions:

1. A dead-end *corridor* shall not be limited in length where the length of the dead-end *corridor* is less than 2.5 times the least width of the dead-end *corridor*.
2. In existing buildings, existing dead-end *corridors* shall be permitted to comply with lengths established in Section 804.8 of the *International Existing Building Code*. Any newly constructed dead-end *corridors* within an existing building shall be limited to the lengths allowed by the *International Building Code*.

C This section references Table 1104.18 for the dead-end corridor limits for existing buildings. The dead-end limit for new construction is in Section 1020.5. Exception 3 in Section 1020.5 matches Exception 1 in this section. With regard to that exception, dead-end corridors are not a concern because travel is not limited to a single path due to the excessive width of the corridor. Exception 2 recognizes existing dead-end corridors, which are permitted by the IEBC but may exceed the limitations set forth in Table 1104.18. This exception would require that any newly created dead-end corridors meet the requirements for new construction. Generally, the dead-end travel limits in Table 1104.18 are more liberal than those in Section 1020.5 for new construction. Note that for dead-end corridors, Table 1104.18 refers to Section 1105.6.6 for Group I-2 occupancies with regard to dead-end corridor limits.

TABLE 1104.18—COMMON PATH, DEAD-END AND TRAVEL DISTANCE LIMITS (by occupancy)

OCCUPANCY	COMMON PATH OF EGRESS TRAVEL LIMIT		DEAD-END LIMIT		EGRESS ACCESS TRAVEL DISTANCE LIMIT	
	Unsprinklered (feet)	Sprinklered (feet)	Unsprinklered (feet)	Sprinklered (feet) ⁱ	Unsprinklered (feet)	Sprinklered (feet)
Group A	75	75 ^j	20 ^a	20 ^a	200	250 ^j
Group B ^h	75 ^g	100 ^j	50	50	200	300 ^j
Group E	75	75 ^j	20	50	200	250 ^j
Group F-1, S-1	75 ^g	100 ^j	50	50	200 ^c	250 ^{c, h, j}
Group F-2, S-2	75 ^g	100 ^j	50	50	300	400 ^j
Group H-1	25	25 ⁱ	0	0	75	75 ^{j, l}
Group H-2	50	100 ^j	0	0	75	100 ^{j, l}
Group H-3	50	100 ^j	20	20	100	150 ^{j, l}
Group H-4	75	75 ⁱ	20	20	150	175 ^{j, l}
Group H-5	75	75 ⁱ	20	50	150	200 ^{j, l}
Group I-1	75	75 ^j	20	50	200	250 ^j
Group I-2	Notes d, e, f	Notes d, e, f, j	Note e	Note e	150	200 ^{b, j}
Group I-3	100	100 ^j	NR	NR	150 ^b	200 ^{b, j}
Group I-4	NR	NR	20	20	200	250 ^j
Group M	75	100 ^j	50	50	200	250 ^{j, j}
Group R-1	75	75 ^{j, k}	50	50	200	250 ^{j, k}
Group R-2	75	125 ^{j, k}	50	50	200	250 ^{j, k}
Group R-3	NR	NR	NR	NR	NR	NR
Group R-4	NR	NR	NR	NR	NR	NR
Group U	75 ^g	100 ^j	20	50	300	400 ^j

NR = No Requirements.

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m².a. See Section 1030.9.5 for dead-end *aisles* in Group A occupancies.

b. This dimension is for the total travel distance, assuming incremental portions have fully utilized their allowable maximums. For travel distance within the room, and from the room exit access door to the exit, see the appropriate occupancy chapter.

c. See Section 412 of the *International Building Code* for special requirements on spacing of doors in aircraft hangars.

d. Separation of exit access doors within a care recipient sleeping room, or any suite that includes care recipient sleeping rooms, shall comply with Section 1105.5.6.

e. In *smoke compartments* containing care recipient sleeping rooms and treatment rooms, dead-end *corridors* shall comply with Section 1105.6.5.

f. In Group I-2, Condition 2, care recipient sleeping rooms or any suite that includes care recipient sleeping rooms shall comply with Section 1105.7.

g. Where a tenant space in Group B, S and U occupancies has an occupant load of not more than 30, the length of a common path of egress travel shall be not more than 100 feet.

h. Where the building, or portion of the building, is limited to one story and the height from the finished floor to the bottom of the ceiling or roof slab or deck is 24 feet or more, the exit access travel distance is increased to 400 feet.

i. For covered and open malls, the exit access travel distance is increased to 400 feet.

j. Buildings equipped with an *approved* automatic sprinkler system in accordance with Section 903.3.1.1.k. Buildings equipped with an *approved* automatic sprinkler system in accordance with Section 903.3.1.2.l. Group H occupancies equipped with an *approved* automatic sprinkler system in accordance with Section 903.2.5.

C This table contains the existing building limits for common path of travel, dead-end limit and travel distance limits. See the commentary to Sections 1104.18, 1104.19 and 1104.20 for a discussion regarding the table.

1104.19 Exit access travel distance. *Exits* shall be located so that the maximum length of *exit access* travel, measured from the most remote point to an *approved exit* along the natural and unobstructed path of egress travel, does not exceed the distances given in Table 1104.18.

C This section references Table 1104.18 for the exit access travel distance limits for existing buildings. The travel distance limits for new construction are in Section 1017. Generally, the travel distance limits in Table 1104.18 for existing construction are the same as those in Table 1017.2 for new construction, except for Group H and I occupancies.

1104.20 Common path of egress travel. The *common path of egress travel* shall not exceed the distances given in Table 1104.18.

C This section refers to Table 1104.18 for the common path of egress travel for existing buildings. The common path of egress travel limits for new construction are in Section 1006.2.1. Generally, the limits in Table 1104.18 for existing construction are the same as in Section 1006.2.1 for new construction.

1104.21 Stairway discharge identification. An *interior exit stairway* or *ramp* that continues below its *level of exit discharge* shall be arranged and marked to make the direction of egress to a *public way* readily identifiable.

Exception: *Stairways* that continue one-half story beyond their *levels of exit discharge* need not be provided with barriers where the *exit discharge* is obvious.

C The requirements of this section are less stringent than the discharge identification requirements for new construction in Section 1023.9. The new construction provisions require that a barrier be placed within the stairway to prevent persons from unintentionally continuing into the levels below the exit discharge. The exception in this section is also not included in the requirements for new construction.

1104.22 Exterior stairway protection. *Exterior exit stairways* shall be separated from the interior of the building as required in Section 1027.6. Openings shall be limited to those necessary for egress from normally occupied spaces.

Exceptions:

1. Separation from the interior of the building is not required for buildings that are two stories or less above grade where the *level of exit discharge* serving such occupancies is the first story above grade.
2. Separation from the interior of the building is not required where the exterior *stairway* is served by an exterior balcony that connects two remote exterior *stairways* or other *approved exits* with a perimeter that is not less than 50 percent open. To be considered open, the opening shall be not less than 50 percent of the height of the enclosing wall, with the top of the opening not less than 7 feet (2134 mm) above the top of the balcony.
3. Separation from the interior of the building is not required for an exterior *stairway* located in a building or structure that is permitted to have unenclosed interior *stairways* in accordance with Section 1023.
4. Separation from the *open-ended corridors* of the building is not required for exterior *stairways* provided that:
 - 4.1. The *open-ended corridors* comply with Section 1020.
 - 4.2. The *open-ended corridors* are connected on each end to an *exterior exit stairway* complying with Section 1027.
 - 4.3. At any location in an *open-ended corridor* where a change of direction exceeding 45 degrees (0.79 rad) occurs, a clear opening of not less than 35 square feet (3 m²) or an exterior *stairway* shall be provided. Where clear openings are provided, they shall be located so as to minimize the accumulation of smoke or toxic gases.

C The exterior stairway requirements of this section are similar to those for new construction in Section 1027. Exception 1 in this section applies to all two-story buildings where the level of the exit discharge is the first story above grade. In new buildings, Group R-1 and R-2 occupancies would not be allowed to apply this exception. Exception 3 is not allowed for new construction. Exception 4 is similar to the allowance given to new construction but does not require the building to be sprinklered or to apply Section 1023.7 for exterior walls.

1104.23 Minimum aisle width. The minimum clear width of *aisles* shall comply with the following:

1. Forty-two inches (1067 mm) for stepped aisles having seating on each side.
Exception: Thirty-six inches (914 mm) where the stepped *aisle* serves fewer than 50 seats.
2. Thirty-six inches (914 mm) for stepped *aisles* having seating on only one side.

Exceptions:

1. Thirty inches (760 mm) for catchment areas serving not more than 60 seats.
2. Twenty-three inches (584 mm) between a stepped aisle *handrail* and seating where a stepped *aisle* does not serve more than five rows on one side.
3. Twenty inches (508 mm) between a stepped *aisle handrail* or guard and seating where the *aisle* is subdivided by a mid-aisle *handrail*.
4. Forty-two inches (1067 mm) for level or ramped *aisles* having seating on both sides.

Exceptions:

1. Thirty-six inches (914 mm) where the *aisle* serves fewer than 50 seats.
2. Thirty inches (760 mm) where the *aisle* serves fewer than 15 seats and does not serve as part of an *accessible route*.
5. Thirty-six inches (914 mm) for level or ramped *aisles* having seating on only one side.
Exception: Thirty inches (760 mm) for catchment areas serving not more than 60 seats and not serving as part of an *accessible route*.
6. In Group I-2, where *aisles* are used for movement of patients in beds, *aisles* shall comply with Section 1105.6.8.

C Items 1 through 5 in this section provide the minimum clear width of aisles for assembly seating. Similar requirements for new construction are in Section 1030.9.1. Several of the exceptions in this section allow for narrower aisles than permitted in Section 1030.9.1; thus, the requirements for existing buildings are less stringent than those for new construction. Item 6 of this section is a pointer to the specific aisle requirements for Group I-2 occupancies where aisles are used for movement of people. See the commentary to Section 1105.6.8.

1104.24 Stairway floor number signs. Existing *stairways* shall be marked in accordance with Section 1023.9.

C This section requires that existing stairs be marked in the same manner as new stairways (see commentary, Section 1023.9). The signage includes important information for people evacuating the building as well emergency responders.

1104.25 Egress path markings. Existing high-rise buildings of Group A, B, E, I, M and R-1 occupancies shall be provided with luminous *egress* path markings in accordance with Section 1025.

Exception: Open, unenclosed stairwells in historic buildings designated as historic under a state or local historic preservation program.

C This section requires that luminous exit path markings be provided in existing buildings on the same basis as in new buildings to facilitate rapid egress and assist in full building evacuation. The text is based on Recommendations 17 and 18 of the National Institute of Standards and Technology's (NIST) report on the World Trade Center (WTC) tragedy. In New York City, after the first bombing of the WTC, requirements were instituted to require exit path markings in vertical exit enclosures in new and existing buildings. This section is taken directly from those requirements.

The marking requirement is only applicable to those buildings having occupied floors more than 75 feet (22 860 mm) above the lowest level of fire department vehicle access. The cost impact on existing buildings is minor when considering the life safety benefit. Therefore, it is logical and affordable to provide existing high-rise buildings with the same protections afforded new high-rise structures.

Historically, code requirements for high-rise buildings were written under the assumption that the building would be evacuated floor by floor. In most instances, in a building equipped throughout with an automatic sprinkler system, only the floor where the fire is located and the floors immediately above and below would be evacuated. Acts of terrorism and accidental incidents, such as power failures, have made it necessary to consider design for full building evacuation that is as rapid as possible. This may be made necessary in response to events both within and outside the building. This provision is intended to facilitate the most rapid possible full building evacuation. See the commentary to Section 1025 for a more complete discussion of luminous egress path markings.

The exception takes into account the aesthetics and possible natural light in an open, unenclosed stairway in a historic building. Also, since the requirement is only applicable to high-rise buildings, the requirement would have limited application in historic structures.

SECTION 1105 — CONSTRUCTION REQUIREMENTS FOR EXISTING GROUP I-2

1105.1 General. Existing Group I-2 shall meet all of the following requirements:

1. The minimum fire safety requirements in Section 1103.
2. The minimum *means of egress* requirements in Section 1104.
3. The additional egress and construction requirements in Section 1105.

Where the provisions of this chapter conflict with the construction requirements that applied at the time of construction, the most restrictive provision shall apply.

C This section provides minimum requirements for existing Group I-2 occupancies. The intent is to increase the minimum safety requirements because of the fragile and sensitive populations within these facilities. These requirements are meant to be applied retroactively. These provisions align with the current approach by the Center for Medicaid and Medicare Services (CMS), the federal authority having jurisdiction. Hospitals are required by CMS to have a life safety survey on a regular basis. If the facility does not meet certain life safety minimums, they are required to upgrade their existing facility. This code change will align the code with those CMS minimum requirements and will hopefully lead to industry consolidation. These retroactive requirements are provided to assist code officials and surveyors during the ongoing regular inspection of hospital facilities and are consistent with the inspections required by federal laws for certification and reimbursement. The requirements reflect consideration of the minimum previously approved construction methods and provide jurisdictions the ability to adopt minimum retroactive provisions that have been vetted by the industry as well as code officials. In addition, these provisions are consistent with current national standards used by the federal government, providing a more uniform level of safety and eliminating many of the current code conflicts for existing facilities.

Several sources were reviewed to determine what the appropriate minimum bar should be, including the current building and fire codes, current CMS guidelines, and previous versions of the model codes. On all issues, enforcement agencies and the regulated facilities weighed in to ensure that these requirements were both necessary and achievable.

This particular section provides scoping for these provisions. Areas in hospitals and nursing homes not in patient care areas will use the general provisions in Sections 1103 and 1104. More restrictive provisions for hospitals or nursing homes are listed in Section 1105. As stated in the general comments at the beginning of this chapter and the commentary to Sections 1101.2 and 1103.5.3, where the buildings must upgrade, consideration of allowable time frames and procedures will need to be worked out between the authority having jurisdiction and those responsible for repairs and alterations in the nursing homes and hospitals.

1105.2 Applicability. The provisions of Sections 1105.3 through 1105.8, 1105.10 and 1105.11 shall apply to the existing Group I-2 *fire area*.

C This section addresses the federal requirement for a separation between Group I-2 occupancies that comply with the requirements in this section and those that do not. Since a building could be several different occupancies, it is reinforcing the need for separation between portions of the building that are compliant with Chapter 11 for Group I-2, and other portions of the buildings. This would require a fire separation between the Group B portion of an existing building that does not comply with all of the minimum retroactive requirements of this chapter.

1105.3 Construction. Group I-2, Condition 2 shall not be located on a floor level higher than the floor level limitation in Table 1105.3 based on the type of construction.

C Section 1105.3 is a retroactive limitation for the allowable height based on construction type because it is a key component of the regulatory approval for a health care facility. While most, if not all, existing hospitals were built to comply with these minimum construction requirements, many were built using methods that predated the current construction type matrix. The allowance for the occupancies, as stipulated in Table 1105.3, are less than that for new construction and are consistent with what is currently mandated for licensing for healthcare facilities.

TABLE 1105.3—FLOOR LEVEL LIMITATIONS FOR GROUP I-2, CONDITION 2

CONSTRUCTION TYPE	AUTOMATIC SPRINKLER SYSTEM	ALLOWABLE FLOOR LEVEL ^a			
		1	2	3	4 or more
IA	Note b	P	P	P	P
	Note c	P	P	P	P
IB	Note b	P	P	P	P
	Note c	P	P	P	P
IIA	Note b	P	P	P	NP
	Note c	P	NP	NP	NP
IIB	Note b	P	P	NP	NP
	Note c	NP	NP	NP	NP
IIIA	Note b	P	P	NP	NP
	Note c	P	NP	NP	NP
IIIB	Note b	P	NP	NP	NP
	Note c	NP	NP	NP	NP
IV	Note b	P	P	NP	NP
	Note c	NP	NP	NP	NP
VA	Note b	P	P	NP	NP
	Note c	NP	NP	NP	NP
VB	Note b	P	NP	NP	NP
	Note c	NP	NP	NP	NP

P = Permitted; NP = Not Permitted.

a. Floor level shall be counted based on the number of stories above grade.

b. The building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1.

c. The building is equipped with an automatic sprinkler system in accordance with Section 1105.8.

1105.4 Incidental uses in existing Group I-2. Incidental uses associated with and located within existing single-occupancy or mixed-occupancy Group I-2 buildings and that generally pose a greater level of risk to such occupancies shall comply with the provisions of Sections 1105.4.1 through 1105.4.3.2.1. Incidental uses in Group I-2 occupancies are limited to those listed in Table 1105.4.

C Incidental use area provisions are applicable to new construction in IBC Section 509; however, similar provisions are needed for existing Group I-2 occupancies since the hazards posed by such rooms or spaces are no different for existing buildings than for new. Section 1105.4 and Table 1105.4 are very similar to and based on IBC Section 509 and Table 509, except that references to occupancies other than Group I-2 are not included. The basic requirements proposed for incidental uses in existing healthcare occupancies rely on the provisions of the IBC for specifics of construction and protection. Sections 1105.4.1 through 1105.4.3.2.1 are based on IBC Sections 509.2 through 509.4.2.1. These provisions will provide correlation with not only the IBC but also with the operational and CMS program standards for existing Group I-2 occupancies.

This specific section establishes the scope of Section 1105.4 and its applicability to Group I-2 occupancies. Incidental uses are rooms or areas that constitute special hazards or risks to life safety that are not typically addressed by the provisions for the occupancy group in which they occur, even though such rooms or areas may functionally be an extension of the primary use. Only those rooms or areas indicated in Table 1105.4 are to be regulated as incidental uses. Incidental uses can be located within both single-occupancy and mixed-occupancy buildings. The concern is that those areas designated as incidental uses pose a risk to the remainder of the building, and as such, some degree of protection is required. The nature of these incidental uses is such that they are small areas not frequented by building occupants, in which a fire could get underway and go unnoticed for a longer time than in a more frequently occupied part of the building.

TABLE 1105.4—INCIDENTAL USES IN EXISTING GROUP I-2 OCCUPANCIES

ROOM OR AREA	SEPARATION AND/OR PROTECTION
Furnace room where any piece of equipment is over 400,000 Btu per hour input	1 hour or provide automatic sprinkler system
Rooms with boilers where the largest piece of equipment is over 15 psi and 10 horsepower	1 hour or provide automatic sprinkler system
Refrigerant machinery room	1 hour or provide automatic sprinkler system
Hydrogen fuel gas rooms, not classified as Group H	2 hours
Incinerator rooms	2 hours and provide automatic sprinkler system
Paint shops not classified as Group H	2 hours; or 1 hour and provide automatic sprinkler system
Laboratories and vocational shops, not classified as Group H	1 hour or provide automatic sprinkler system
Laundry rooms over 100 square feet	1 hour or provide automatic sprinkler system
Patient rooms equipped with padded surfaces	1 hour or provide automatic sprinkler system
Physical plant maintenance shops	1 hour or provide automatic sprinkler system
Waste and linen collection rooms with containers with total volume of 10 cubic feet or greater	1 hour or provide automatic sprinkler system
Storage rooms greater than 100 square feet	1 hour or provide automatic sprinkler system
Stationary storage battery systems having a liquid electrolyte capacity of more than 50 gallons for flooded lead-acid, nickel cadmium or VRLA, or more than 1,000 pounds for lithium-ion and lithium metal polymer used for facility standby power, emergency power or uninterruptible power supplies	2 hours
For SI: 1 square foot = 0.0929 m ² , 1 pound per square inch (psi) = 6.9 kPa, 1 British thermal unit (Btu) per hour = 0.293 watts, 1 horsepower = 746 watts, 1 gallon = 3.785 L.	

1105.4.1 Occupancy classification. Incidental uses shall not be individually classified in accordance with Section 302.1 of the *International Building Code*. Incidental uses shall be included in the building occupancies within which they are located.

C Consistent with the IBC, this section expressly states that incidental uses are not considered as separate and distinct occupancy classifications but, rather, are classified the same as the occupancies in which they are located. As an example, a waste and linen collection room in a hospital would be classified as a portion of the Group I-2 occupancy even though it may present a level of hazard more akin to a Group S-1 occupancy.

1105.4.2 Area limitations. Incidental uses shall not occupy more than 10 percent of the *building area* of the story in which they are located.

C The proposed floor area limitation of 10 percent for incidental uses emphasizes the ancillary nature of such rooms and areas and correlates with the IBC. Each incidental use would be limited to a maximum floor area of 10 percent of the floor area of the story in which it is located. Where there are two or more tenants located on the same story, the 10 percent limitation is based on the floor area of each individual tenant space rather than that of the entire story. The application of the limit on a tenant-by-tenant basis is consistent with the concept of incidental uses typically being ancillary only to a portion of the building (i.e., the specific tenant occupancy).

1105.4.3 Separation and protection. The incidental uses listed in Table 1105.4 shall be separated from the remainder of the building or equipped with an *automatic sprinkler system*, or both, in accordance with the provisions of that table.

C In addition to identifying those rooms or areas that warrant regulation as incidental uses, Table 1105.4 also indicates the required degree of protection or separation. The requirements identified in Table 1105.4 vary, depending on the incidental use. In some cases, a specific type of separation and/or protection is required, while in others there is an option.

1105.4.3.1 Separation. Where Table 1105.4 specifies a *fire-resistance-rated* separation, the incidental uses shall be separated from the remainder of the building in accordance with Section 509.4.1 of the *International Building Code*.

C Where a fire-resistance rated separation is required, the incidental use must be separated from other portions of the building in accordance with assemblies complying with the IBC.

1105.4.3.2 Protection. Where Table 1105.4 permits an *automatic sprinkler system* without a *fire-resistance-rated* separation, the incidental uses shall be separated from the remainder of the building by construction capable of resisting the passage of smoke in accordance with Section 509.4.2 of the *International Building Code*.

C In this section, where Table 1105.4 allows protection by an automatic sprinkler system without a fire-resistance-rated separation, the construction enclosing the incidental use still must resist the passage of smoke. Construction details for resisting the passage of smoke are provided in the IBC. Note that this requirement is not specifying a smoke partition. The smoke partition requirements in IBC Section 710 are specific requirements that must be directly referenced.

1105.4.3.2.1 Protection limitation. Except as otherwise specified in Table 1105.4 for certain incidental uses, where an *automatic sprinkler system* is provided in accordance with Table 1105.4, only the space occupied by the incidental use need be equipped with such a system.

C This section makes it clear that the sprinkler systems stipulated in Table 1105.4 would be required for the incidental use area only.

1105.5 Corridor construction. In Group I-2, in areas housing patient sleeping or care rooms, *corridor* walls and the opening protectives therein shall provide a barrier designed to resist the passage of smoke in accordance with Sections 1105.5.1 through 1105.5.7.

C The focus of this section is specifically for corridor construction in patient care and sleeping room areas in Group I-2 occupancies. The construction of corridors is intended to provide protection that will resist the passage of smoke. The provisions address the type of materials required, fire-resistance-rating requirements, opening protection and corridor wall continuity.

Existing corridor construction should primarily be evaluated for its ability to resist or limit the transfer of smoke, regardless of the code at the time of construction. This is consistent with the defend-in-place scenario for patient protection during a fire event (see Section 405). Corridor walls, even if they were built 60 years ago, should be regularly assessed to confirm that they minimize the transfer of smoke. This section describes some criteria by which this can be assessed.

1105.5.1 Materials. The walls shall be of materials permitted by the building type of construction.

C This section simply requires that the materials used to construct the corridors be consistent with the type of construction. If it is a noncombustible construction (Type I or II), the walls must be constructed in accordance with IBC Table 601 unless IBC Section 603 provides any exceptions.

1105.5.2 Fire-resistance rating. Unless required elsewhere in this code, *corridor* walls are not required to have a *fire-resistance rating*. *Corridor* walls that were installed as *fire-resistance-rated* assemblies in accordance with the applicable codes under which the building was constructed, remodeled or altered shall be maintained unless modified in accordance with the *International Existing Building Code*.

C The key with corridor walls is more about the resistance of the passage of smoke and less about the fire resistance. This section clarifies that these walls are not required to be fire-resistance rated unless another requirement, such as for a smoke barrier or incidental use area separation, would prompt such fire-resistance rating. Also, if the corridor was originally constructed as a fire-resistance-rated assembly as required at the time it was constructed, those corridor walls must be maintained unless the IEBC allows a reduction. IEBC Sections 501.2 and 802.6 allow a reduction in fire-resistance-rated assemblies based on an overall analysis of the building. The concept is that once a building without sprinkler protection has been sprinklered throughout, whether due to renovations or retroactive code application, the designer should be permitted to take advantage of some of the sprinkler trade-offs that are allowed for new construction, such as the allowance for healthcare corridors to be smoke partitions instead of needing to be fire partitions. Corridors in I-2 occupancies were required to be fire partitions for decades in nonsprinklered hospitals, and also in sprinklered hospitals built to one of the legacy codes. A proper review by the building code official must be performed to ensure that such reductions are consistent with the level of safety provided by the current building code. This proposal attempts to provide a mechanism for that process by adding a reference to the IEBC.

It should be noted that, according to the NFPA “Report on Fires in Health Care Facilities” published November 2012, between 2006 and 2010, sprinklers were present in only 55 percent of reported health care fires. Although those statistics may not precisely gauge the exact proportion of healthcare facilities without sprinklers today, the fact remains that a substantial number of existing Group I-2 occupancies are not sprinklered throughout. Revising this code article to lay out a clear path for reducing the required fire-resistance rating of corridors can only assist in incentivizing older hospitals to have sprinklers retrofitted as soon as possible. See also the commentary to Sections 1103.5.2, 1103.5.3 and 1105.9.

1105.5.3 Corridor wall continuity. *Corridor* walls shall extend from the top of the foundation or floor below to one of the following:

1. The underside of the floor or roof sheathing, deck or slab above.
2. The underside of a ceiling above where the ceiling membrane is constructed to limit the passage of smoke.
3. The underside of a lay-in ceiling system where the ceiling system is constructed to limit the passage of smoke and where the ceiling tiles weigh not less than 1 pound per square foot (4.88 kg/m²) of tile.

C As discussed in Section 1105.5.2, providing a barrier is the key to corridor effectiveness. Section 1105.5.2 does not require a fire-resistance rating for existing corridors, but the separation that the corridor provides must be continuous to a point where the separated spaces are protected. This section provides three solutions to achieve this. The first focuses simply on extending all the way to the floor or roof sheathing, deck or slab above. The other options allow the separation to stop at the finished ceiling. Ending the separation at the ceiling membrane is conditioned on the ceiling being able to resist the passage of smoke. In the case of lay-in ceilings, they must also meet the minimum weight requirement to increase the likelihood that they will stay in place.

1105.5.4 Openings in corridor walls. Openings in *corridor* walls shall provide protection in accordance with Sections 1105.5.4.1 through 1105.5.4.3.

C Openings in corridor walls present the largest failures of such separations. This section addresses both windows and doors located in corridor walls. Again, these provisions are intended to address the condition of existing walls and may not be consistent with new construction.

1105.5.4.1 Windows. Windows in *corridor* walls shall be sealed to limit the passage of smoke, or the window shall be automatic-closing upon detection of smoke, or the window opening shall be protected by an automatic closing device that closes upon detection of smoke.

Exception: In *smoke compartments* not containing patient sleeping rooms, pass-through windows or similar openings shall be permitted in accordance with Section 1105.5.4.3.

C Windows must effectively employ one of the following three methods to resist the passage of smoke:

- Proper sealing.
- Automatic closing on detection of smoke (smoke detection system).
- Automatic closing by automatic closing device (single-station smoke detector).

The exception addresses lower risk areas where no patient sleeping rooms are present.

1105.5.4.2 Doors. Doors in *corridor* walls shall comply with Sections 1105.5.4.2.1 through 1105.5.4.2.3.

C This section addresses various different issues related to doors in corridor walls. These provisions provide some basic safety while providing flexibility in existing hospitals. As with the other provisions, these requirements and allowances are consistent with federal compliance requirements.

1105.5.4.2.1 Louvers. Doors in *corridor* walls shall not include louvers, transfer grills or similar openings.

Exception: Doors shall be permitted to have louvers, transfer grills or similar openings at toilet rooms or bathrooms; storage rooms that do not contain storage of flammable or combustible material; and storage rooms that are not required to be separated as incidental uses.

C Louvers, transfer grilles or similar openings present an increased risk of smoke passing into protected spaces. Therefore, these openings are generally prohibited. The exception addresses spaces where passage of smoke presents a reduced risk as such spaces are not typically occupied or their occupancy is limited.

1105.5.4.2.2 Corridor doors. Doors in *corridor* walls shall limit the transfer of smoke by complying with the following:

1. Doors shall be constructed of not less than 1³/₄ inch-thick (44 mm) solid bonded-core wood or capable of resisting fire not less than 1¹/₃ hour.
Exception: *Corridor* doors in buildings equipped throughout with an *automatic sprinkler system*.
2. Frames for side-hinged swinging doors shall have stops on the sides and top to limit transfer of smoke.
3. Where provided, vision panels in doors shall be a fixed glass window assembly installed to limit the passage of smoke. Existing wired glass panels with steel frames shall be permitted to remain in place.
4. The clearance between the bottom of the door and floor shall not exceed 1 inch (25 mm).
5. Doors shall be positive latching with devices that resist not less than 5 pounds (22.2 N). Roller latches are prohibited.
6. Mail slots or similar openings shall be permitted in accordance with Section 1105.5.4.3.

C This section provides the list of necessary requirements that existing doors in corridors must meet. Often the criteria provide some allowances not given for new construction. All criteria must be addressed if applicable.

Item 1 requires a door to be of substantial construction in nonsprinklered buildings. Either a 1/3-hour fire-rated door is required or an equivalent door is prescribed. The exception allows no rating for the door in buildings equipped throughout with an automatic sprinkler system.

Item 2 requires frames to have stops on the sides and top to minimize the transfer of smoke on swinging doors.

Item 3 addresses vision panels. This item allows wired glass panels in steel frames to continue to be used where they would not be allowed in new construction.

Item 4 minimizes the clearance under the door to 1 inch to prevent additional smoke passage.

Item 5 addresses how such doors are required to latch. Roller latches are prohibited as they may be compromised during a fire event and lead to the door opening.

Item 6 is consistent with Section 1105.5.4.3 and allows mail slots or similar openings under certain conditions for doors in corridors.

1105.5.4.2.3 Dutch doors. Where provided, Dutch doors shall comply with Section 1105.5.4.2.2. In addition, Dutch doors shall be equipped with latching devices on either the top or bottom leaf to allow leaves to latch together. The space between the leaves shall be protected with devices such as astragals to limit the passage of smoke.

C Dutch doors have been used in health care facilities for many years for various necessary operational reasons. While the IBC does not specifically speak of Dutch doors, their use is not prohibited but must meet the requirements contained in IBC Section 407.3, including positive latching and limiting the transfer of smoke. This section provides clarity for existing installations by giving specific guidance on the minimum acceptable requirements, including positive latching and smoke transfer, for their use in corridor walls. A definition is provided in Section 202 for further clarity.

1105.5.4.2.4 Self- or automatic-closing doors. Where self- or automatic-closing doors are required, closers shall be maintained in operational condition. Hold-open devices on doors shall be capable of manual release.

C This section is provided to make sure that self- or automatic-closing doors continue to be operational. If automatic-closing doors that are no longer operational are propped open, the function of the corridor is reduced. It is important to note that this section does not require self-closing or automatic-closing doors; it simply requires that they be maintained. The last sentence addresses the operation of hold-open devices often used on corridor doors. This simply clarifies that hold-open devices need to be capable of being released both manually and via an electrical or mechanical system operation.

1105.5.4.2.5 Protective plates. Protective plates installed on *corridor* doors shall not be limited in size.

C This section provides direction that protective plates attached to doors to protect the viability of the door are not limited in size. This is a common attachment on doors. This section reduces confusion as to whether such attachments are permitted and how they might affect the listing.

1105.5.4.3 Openings in corridor walls and doors. In other than *smoke compartments* containing patient sleeping rooms, mail slots, pass-through windows or similar openings shall not be required to be protected where the aggregate area of the openings between the *corridor* and a room are not greater than 80 square inches (51 613 mm²) and are located with the top edge of any opening not higher than 48 inches above the floor.

C This section is essentially an exception for existing mail slot, pass-through and similar openings that are commonly found in hospitals. These are needed for privacy, medication security and other operational needs. The restrictions on their location are consistent with federal requirements.

1105.5.5 Penetrations. The space around penetrating items shall be filled with an *approved* material to limit the passage of smoke.

C Penetrations, though smaller than door and window openings, can create locations for smoke to pass through and must be addressed. Group I-2 occupancies often have many pipes penetrating walls that must be addressed. Many smaller openings can lead to a larger overall area of smoke passage.

1105.5.6 Joints. Joints shall be filled with an *approved* material to limit the passage of smoke.

C Similar to penetrations, joints that are formed at the intersections of walls and floor/ceiling assemblies and roof assemblies can create other avenues of smoke passage. If not addressed, these small openings can lead to a hazard during building fires.

1105.5.7 Ducts and air transfer openings. The space around a duct penetrating a smoke partition shall be filled with an *approved* material to limit the passage of smoke. Air transfer openings in smoke partitions shall be provided with a smoke damper complying with Section 717.3.2.2 of the *International Building Code*.

Exception: Where the installation of a smoke damper will interfere with the operation of a required smoke control system in accordance with Section 909, *approved* alternative protection shall be utilized.

C Anytime a duct penetrates a wall, the space around the duct must be minimally protected to prevent smoke passage during fire. Group I-2 occupancies may have complex HVAC systems that can lead to a multitude of ducts passing through walls. Air transfer openings through corridor walls need to be protected with smoke dampers in accordance with IBC Section 717.3.2.2 unless such dampers will affect the performance of a smoke control system.

1105.6 Means of egress. In addition to the *means of egress* requirements in Section 1104, Group I-2 facilities shall meet the *means of egress* requirements in Sections 1105.6.1 through 1105.6.8.

C These provisions are intended to provide minimal means of egress requirements specific to Group I-2 occupancies, where these occupancies work on a defend-in-place strategy and the needs are unique. Sections 1105.6.1 through 1105.6.8 address a variety of issues, such door size, corridor width, dead ends and exit separation.

1105.6.1 Two means of egress. A *means of egress* shall be provided from each *smoke compartment* created by *smoke barriers* without having to return through the *smoke compartment* from which the *means of egress* originated. *Smoke compartments* that do not contain an *exit* shall be provided with direct access to not less than two adjacent *smoke compartments*.

C This section aligns the requirement for exits within existing Group I-2 smoke compartments with the building code. Repeating the language establishes the criteria preferred for existing facilities as an independent requirement from new construction. If the requirement changes in the future for new construction, this requirement should remain for existing buildings.

1105.6.2 Size of door. *Means of egress* doors used for the movement of patients in beds shall provide a minimum clear width of 41½ inches (1054 mm). The height of the door opening shall be not less than 80 inches (2032 mm).

Exceptions:

1. Door closers and door stops shall be permitted to be 78 inches (1981 mm) minimum above the floor.
2. In Group I-2, Condition 1, existing *means of egress* doors used for the movement of patients in beds that provide a minimum clear width of 32 inches (813 mm) shall be permitted to remain.

C This section is consistent with Section 1104.7. It follows the format and requirements of Section 1010.1.1. There are two exceptions that allow the door closers and door stops to not meet the door-opening height. This is a reasonable allowance for existing buildings and is consistent with Exception 4 to Section 1010.1.1, which allows door-opening heights in dwelling units and sleeping units to be 78 inches. The second exception provides flexibility to Group I-2, Condition 1 occupancies such as nursing homes, which do not depend on these door widths being available as frequently for bed movement.

1105.6.3 Group I-2 occupancies. In Group I-2, where a door serves as an opening protective in a *fire barrier*, *smoke barrier* or *fire wall* and where the door is equipped with a hold-open device, such door shall automatically close upon any of the following conditions:

1. Actuation of smoke detectors initiating the hold-open device.
2. Activation of the *fire alarm system* within the zone.
3. Activation of an *automatic sprinkler system* within the zone.

C This section allows doors to be held open with specific requirements as to how they are required to be released. Without this language, it was unclear whether these mechanisms would be permitted to allow doors to remain open on a regular basis. There was concern that the existing provisions in the IFC could be more restrictive than the IBC with regard allowing hold-open devices. This allowance of hold-open devices prevents damage to the door when a stretcher or supply cart moves through the barrier, providing significant operational savings by increasing the useful life of the rated door.

1105.6.4 Ramps. In areas where *ramps* are used for movement of patients in beds, the clear width of the *ramp* shall be not less than 48 inches (1219 mm).

C This section, which is referenced from Section 1104.15, provides the width necessary for ramps in Group I-2 occupancies where beds are being moved. Section 1104.15 only requires a 30-inch width.

1105.6.5 Corridor width. In areas where *corridors* are used for movement of patients in beds, the clear width of the *corridor* shall be not less than 48 inches (1219 mm).

C This section is consistent with the width required for ramps in Section 1105.6.4 and is focused on the width required for the movement of beds. The requirement of 48 inches clear width is less restrictive than that required of new construction in Section 1020.3. New Group I-2 occupancies require a 96-inch clear width.

1105.6.6 Dead-end corridors. In *smoke compartments* containing patient sleeping rooms and treatment rooms, dead-end *corridors* shall not exceed 30 feet (9144 mm) unless *approved* by the *fire code official*.

C The dead-end corridor requirements are specific to Group I-2 occupancies. Typically, in new construction the dead-end corridor would be limited to 20 feet. This provides more flexibility but is consistent with the CMS requirements.

1105.6.7 Separation of exit access doors. Patient sleeping rooms, or any suite that includes patient sleeping rooms, of more than 1,000 square feet (92.9 m²) shall have not less than two *exit access* doors placed a distance apart equal to not less than one-third of the length of the maximum overall diagonal dimension of the patient sleeping room or suite to be served, measured in a straight line between *exit access* doors.

C This requirement addresses the need for exit separation in larger patient sleeping rooms and suites. Once they become large enough, the need for two exits becomes more important. This section is less restrictive than Section 1007.1.1 by only requiring one-third the diagonal of the space versus one-half. This requirement may result in the need for an additional exit access door in such spaces.

1105.6.8 Aisles. In areas where *aisles* are used for movement of patients in beds, the clear width of the *aisle* shall be not less than 48 inches (1219 mm).

C This requirement is consistent with the widths established for ramps and corridors in Sections 1105.6.4 and 1105.6.5, respectively. Again, this width is associated with areas where bed movement will occur.

1105.7 Smoke compartments. *Smoke compartments* shall be provided in existing Group I-2, Condition 2, in accordance with Sections 1105.7.1 through 1105.7.6.

C Smoke compartments, used in healthcare facilities to limit the movement of smoke, are a key component of the defend-in-place strategy, a strategy where occupants are protected from fire without relocation. These compartments act as safe locations for patients by preventing the spread of smoke. Through compartmentalization, patients may remain safely in their rooms as fire suppression systems and fire responders extinguish the fire. Under severe fire conditions that threaten the immediate compartment area, patients may be evacuated horizontally to the safety of an adjacent compartment on the same floor. Being able to do this is critical since, because of the health status of many patients, their evacuation from the building might put them in danger. The proper design, construction and application of smoke compartments will provide added protection, buy valuable time and save lives of critically ill patients before a total evacuation may become necessary.

These requirements are provided to assist code officials and surveyors during the ongoing regular inspection of hospital facilities. These inspections are required by federal laws for certification and reimbursement. These requirements consider the minimum previously approved construction methods. This is consistent with the federal requirements to which these facilities are currently held. Sections 1105.7.1 through 1105.7.6 provide all the necessary elements for existing facilities.

1105.7.1 Design. *Smoke barriers* shall be provided to subdivide each story used for patients sleeping with an *occupant load* of more than 30 patients into not fewer than two *smoke compartments*.

C The defend-in-place concept is a basic minimum level of safety for these facilities. Every facility should be equipped with at least two smoke compartments for temporary relocation of patients. This section addresses acceptable configurations of smoke barrier walls and smoke barriers in existing hospitals for each story with sleeping rooms accommodating 30 or more patients.

1105.7.1.1 Refuge areas. Refuge areas shall be provided within each *smoke compartment*. The size of the refuge area shall accommodate the occupants and care recipients from the adjoining *smoke compartment*. Where a *smoke compartment* is adjoined by two or more *smoke compartments*, the minimum area of the refuge area shall accommodate the largest *occupant load* of the adjoining compartments.

The size of the refuge area shall provide the following:

1. Not less than 30 net square feet (2.8 m²) for each care recipient confined to a bed or stretcher.
2. Not less than 15 square feet (1.4 m²) for each resident in a Group I-2 using mobility assistance devices.
3. Not less than 6 square feet (0.56 m²) for each occupant not addressed in Items 1 and 2.

Areas of spaces permitted to be included in the calculation of the refuge area are *corridors*, sleeping areas, treatment rooms, lounge or dining areas and other low-hazard areas.

C This section addresses adequate sizing of refuge areas. IBC Section 407.5.3 also includes requirements for sizing refuge areas. Otherwise, the provisions are fairly consistent.

1105.7.2 Smoke barriers. *Smoke barriers* shall be constructed in accordance with Section 709 of the *International Building Code*.

Exceptions:

1. Existing *smoke barriers* are permitted to remain where the existing *smoke barrier* has a minimum *fire-resistance rating* of $\frac{1}{2}$ hour.
2. *Smoke barriers* shall be permitted to terminate at an atrium enclosure in accordance with Section 404.6 of the *International Building Code*.

C Generally, compliance with the IBC is required for smoke barriers. However, the first exception is intended to bring non-compliant smoke barriers to at least a 1/2-hour fire-resistance rating. Previously approved smoke barriers are not intended to be reduced to 1/2 hour. Chapter 7 requires maintenance of approved construction. The second exception addresses termination of the smoke barrier. Typically, smoke barriers are required to terminate at outside walls.

1105.7.3 Opening protectives. Openings in *smoke barriers* shall be protected in accordance with Section 716 of the *International Building Code*. Opening protectives shall have a minimum *fire protection rating* of $\frac{1}{3}$ hour.

Exceptions:

1. Existing wired glass vision panels in doors shall be permitted to remain.
2. Existing nonlabeled protection plates shall be permitted to remain.

C This section addresses doors in smoke barriers in existing Group I-2 occupancies. This section requires compliance as applicable with Section 716. The rating of doors is required to be a minimum of 20 minutes. The first exception allows openings in doors with wired glass to remain. The second exception correlates with IBC Section 709.5, Exception 1. Smoke barrier

doors are typically installed across corridors and patient treatment areas. These doors see a very high volume of gurney and bed traffic, as well as carts, wheeled equipment and transport devices. As a result, they are often damaged. Exception 2 allows the installation of a nonlabeled protective plate, usually made of steel or other resilient material, to be installed on these doors to protect them from excessive wear and damage. Due to the size of equipment being wheeled through, these protective plates need to be allowed to be greater than 48 inches high. Currently NFPA 80 would require that the protective plates on rated doors be limited to 48 inches and that they be labeled. The doors in smoke barriers do not function as true fire doors. This section contains many special directives and requirements exempting smoke barrier doors from meeting fire door requirements. Smoke barriers are intended to be substantial construction. Protective plates provide additional protection to the doors, keeping the original construction free from damage while creating a more substantial door. They do not provide the same fire-resistance rating as a true 1-hour fire barrier.

1105.7.4 Penetrations. Penetrations of *smoke barriers* shall comply with the *International Building Code*.

Exception: *Approved* existing materials and methods of construction.

C Penetrations, as with corridor construction, can be numerous and lead to smoke movement from one smoke compartment to another. This section requires that such penetrations be dealt with in accordance with the IBC. Sealing penetrations is a reasonable requirement to increase patient safety. There is some flexibility provided if such penetrations have already been addressed to the satisfaction of the fire code official.

1105.7.5 Joints. Joints made in or between *smoke barriers* shall comply with the *International Building Code*.

Exception: *Approved* existing materials and methods of construction.

C Joints formed by the intersection of smoke barriers and floor/ceiling and roof assemblies can often create gaps where smoke can spread. This simply requires compliance with the IBC to protect those joints. Similar to Section 1105.7.4 addressing penetrations, there is some flexibility provided if such joints have already been addressed to the satisfaction of the fire code official.

1105.7.6 Duct and air transfer openings. Penetrations in a *smoke barrier* by duct and air transfer openings shall comply with Section 717 of the *International Building Code*.

Exception: Where existing duct and air transfer openings in *smoke barriers* exist without smoke dampers, they shall be permitted to remain. Any changes to existing smoke dampers shall be submitted for review and *approved* in accordance with Section 717 of the *International Building Code*.

C Similar to penetrations and joints, where ducts and air transfer openings penetrate a smoke barrier they must be sealed in accordance with the IBC. IBC Section 717.5.5 requires a smoke damper unless Exception 2 to that section is met. Exception 2 to Section 717.5.5 recognizes a fully ducted system in a building equipped throughout with an automatic sprinkler system. See the commentary to IBC Section 717.5.5. The application of that exception is limited to Group I-2, Condition 2 occupancies.

The exception to this section recognizes that existing duct and air transfer openings may be permitted to remain without smoke dampers. Any modification of existing smoke dampers would have to go through the normal process for making an alteration to existing construction. The exception is not intended to allow the removal of existing smoke dampers but instead allow the continuance of existing conditions that did not require them.

1105.8 Group I-2 care suites. Care suites in existing Group I-2, Condition 2 occupancies shall comply with Sections 407.4.4 through 407.4.4.7.2 of the *International Building Code*.

C This proposal defines the requirements for care suites (both sleeping and nonsleeping) that are an integral design concept for many areas within a hospital. Typical uses include intensive care units, operating rooms, emergency departments and imaging departments. The suites allow for better and safer care than nonsuite options. These provisions deal with common paths of travel, separation of exit access doors, and number of doors passed through (i.e., previously intervening rooms) in suites. See the commentary to IBC Sections 407.4.4 through 407.4.4.7.2.

1105.9 Group I-2 automatic sprinkler system. An *automatic sprinkler system* installed in accordance with Section 903.3.1.1 shall be provided throughout the floor containing the Group I-2 *fire area*. The sprinkler system shall be provided throughout the floor where the Group I-2 occupancy is located, on all floors between the Group I-2 occupancy *fire area* and the *level of exit discharge*, the *level of exit discharge*, and all floors below the *level of exit discharge*.

Exception: Floors classified as an open parking garage are not required to be sprinklered.

C This section requires that an automatic sprinkler system be provided on the floor containing the Group I-2 occupancy, all floors between that level and the level of exit discharge, and all floors below the level of exit discharge. This is a general requirement for all Group I-2 occupancies. Automatic sprinkler protection of existing Group I-2 occupancies is of critical importance due to the nature of the occupants. In Group I-2 occupancies, the occupants are, by definition, considered not capable of self-preservation. The evacuation difficulties associated with these occupants create the need to incorporate a defend-in-place philosophy of fire protection in occupancies in Group I-2. For this reason, all such existing occupancies are to be protected with an automatic sprinkler system.

This is an especially important requirement for nursing homes. An NFPA report, *Facilities that Care for the Aged Including Nursing Homes and Residential Board and Care*, states, “The death rate per 1,000 fires was 82 percent lower when automatic suppression systems were present.” The report further states, “Residents of these facilities are particularly vulnerable. The risk increases with increasing age. Consequently, the aged are considered a high-risk population. Institutional facilities that care for older adults must work diligently to prevent fires and to train staff and to equip the property (e.g., active systems) for effective response should a fire occur.”

Note that Section 1103.5.3 requires that an automatic sprinkler system be installed throughout the entire building containing a Group I-2, Condition 2 occupancy. See the commentary to Section 1103.5.3 for more discussion on the intended application of that section.

1105.10 Group I-2 automatic fire alarm system. An automatic *fire alarm system* shall be installed in existing Group I-2 occupancies in accordance with Section 907.2.6.2.

Exception: Manual fire alarm boxes in patient sleeping areas shall not be required at *exits* if located at all nurses’ control stations or other constantly attended staff locations, provided such that manual fire alarm boxes are visible, are provided with *ready access*, and travel distances required in Section 907.4.2.1 are not exceeded.

C Because care recipients may be incapable of self-evacuation and thus would rely on staff for assistance, an approved fire alarm system is required in existing Group I-2 occupancies. The system must comply with Section 907.2.6.2 (see commentary, Section 907.2.6.2). Note that Section 1105.9 requires sprinkler protection in existing Group I-2 occupancies and that the requirements of this section are in addition thereto. This provision is not intended to override the allowances for “private mode” in Section 907.2.6.2. Typical audible and visible alarms could be a detriment in some areas of a hospital, such as where patients are undergoing procedures.

1105.11 Essential electrical systems. Essential electrical systems in Group I-2, Condition 2 occupancies shall be in accordance with Sections 1105.11.1 and 1105.11.2.

C Both requirements in Sections 1105.11.1 and 1105.11.2 are specific to Group I-2, Condition 2 occupancies. This section addresses the requirements related to the needs of essential power and the reassessment to ensure that current installations will still meet the needs of the hospitals they support.

1105.11.1 Where required. Where required by NFPA 99, Group I-2, Condition 2 occupancies shall be provided with an essential electrical system in accordance with NFPA 99.

C This section requires existing Group I-2, Condition 2 occupancies to provide essential electrical systems where NFPA 99 would require such systems. NFPA 99 provides a risk-based approach to determine if essential power supplies are necessary. This analysis looks at the risk of failure of the system and its implications to the patients or caregivers in terms of injury or death. Specific design considerations are set based on the risk.

1105.11.2 Installation and duration. In Group I-2, Condition 2 occupancies, the installation and duration of operation of existing essential electrical systems shall be based on a hazard vulnerability analysis conducted in accordance with NFPA 99.

C This section requires a reassessment of the essential electrical systems based on a hazard vulnerability analysis to make sure that the systems meet the needs of the facilities for emergencies. This requirement would apply whether or not the facility is being altered or has been damaged. If “substantial improvement” of the building occurs, per the IBC and IEBC definition, ASCE 24 would require this analysis. The flood provision section addresses such concerns as the location of equipment in a flood plain.

SECTION 1106—REQUIREMENTS FOR OUTDOOR OPERATIONS

1106.1 Tire storage yards. Existing tire storage yards shall be provided with fire apparatus access roads in accordance with Sections 1106.1.1 and 1106.1.2.

C This section introduces the fire apparatus access provisions that follow.

1106.1.1 Access to piles. Access roadways shall be within 150 feet (45 720 mm) of any point in the storage yard where storage piles are located not less than 20 feet (6096 mm) from any storage pile.

C See the commentary to Section 3406.2.

1106.1.2 Location within piles. Fire apparatus access roads shall be located within all pile clearances identified in Section 3405.4 and within all fire breaks required in Section 3405.5.

C See the commentary to Sections 3405.4 and 3405.5.

SECTION 1107—ENERGY STORAGE SYSTEMS

1107.1 Lithium-ion technology energy storage systems. The owner of an energy storage system (ESS) utilizing lithium-ion battery technology having capacities exceeding the values in Table 1207.1.3 and installed prior to the jurisdiction’s adoption of the 2018 or

later edition of the *International Fire Code* shall provide the *fire code official* a failure modes and effects analysis (FMEA) or other approved hazard mitigation analysis in accordance with Section 104.2.2 for review and approval.

Exception: Detached one- and two-family dwellings and townhouses.

C This section addresses the rapidly evolving technology associated with energy storage systems (ESS) using lithium-ion batteries and is intended to fill the gap between existing and new installations. Many of the installations were constructed prior to the provisions found in this code. Due to risks associated with lithium-ion ESS, this section requires an analysis of potential hazards.

1107.1.1 Early detection. In addition to the requirements of Sections 1207.1.8.1 and 1207.1.8.2, the analysis shall include an assessment of the ability of the installed protection systems to provide for early detection and notification of a thermal runaway event in relation to the ability of emergency responders to safely mitigate the size and impact of a thermal runaway event.

C The failure modes and effects analysis (FMEA) or other approved hazard mitigation analysis must evaluate the ability of the installed protection systems to provide early detection and notification to emergency responders of thermal runaway so that they may safely mitigate the impact of a thermal runaway.

Planning in advance of an incident, including a thorough assessment of fire and emergency service capabilities, must be part of the analysis.

1107.1.2 Corrective action plan. Where hazards are identified by the analysis, a plan that includes a timetable for corrective action shall be submitted to the *fire code official* for review and approval. The plan shall include actions and system improvements necessary for eliminating or mitigating any identified hazards, including listed methods for early detection and notification of a thermal runaway event.

C Should the failure modes and effects analysis (FMEA) or other hazard mitigation analysis find deficiencies in the lithium-ion energy storage system, the owner must submit to the fire code official a timetable for correction. The plan must include actions and system improvements necessary for eliminating or mitigating any identified hazards, including listed methods for early detection and notification of a thermal runaway.

The fire code official may insist the permit applicant retain a qualified specialist, engineer or firm to evaluate the report and recommendations and provide technical assistance at no cost to the jurisdiction in accordance with Section 104.2.2.

Bibliography

The following resource materials were used in the preparation of the commentary for this chapter of the code:

Ahrens, Marty. *Facilities that Care for the Aged Including Nursing Homes and Residential Board and Care*. Quincy, MA: National Fire Protection Association, 2006.